Stearley Heights Admin Conversion Study
Kadena Air Base, Japan

Conceptual Charrette Report

Contract No. W912BV-19-D-0008
Task Order No. W912BV-23-F-0057

Final Submittal

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1 Introduction

1.1 Purpose of Conceptual Charrette Report

This Conceptual Charrette Report (CCR) documents the validation process for the Stearley Heights Admin Conversion Study at Kadena Air Base (AB), Japan. Jacobs was contracted by the U.S. Army Corps of Engineers (USACE), Tulsa District, under Contract W912BV-19-D-0008, Task Order W912BV-23-F-0057, to validate the scope, requirements, and cost for the project. The objective of the effort is to clearly define and confirm project requirements, scope, cost, and criteria. It also provides the conceptual framework for executing the project as required by the user. Active involvement by the building users during all phases of design and construction is essential to ensure the facility meets all functional requirements. The Design Agent (DA) is responsible for verifying the accuracy of the information contained within this document.

The Jacobs team first reviewed the statement of work provided in the request for proposal and, in April 2023, the task was awarded. The project stakeholders attended a monthly teleconference meeting for the Kadena AB task orders until the planning charrette for the Stearley Heights Admin Conversion Study was scheduled. Additional project information was obtained from the 718th Civil Engineering Squadron (718 CES) prior to the planning charrette, which was held on 16 to 18 January 2024 at Kadena AB. The out-brief for the Admin Conversion Study occurred on 18 January 2024 at Kadena AB to inform the stakeholders of the general condition of the facility and to present three courses of action (COAs). From the three COAs, a Preferred COA was selected and was used as the basis for this report, which documents the findings and subsequent analysis of requirements to bring the facility up to building code standards and meet the requirements set forth by the Kadena AB stakeholders.

1.2 Project Overview

1.2.1 Purpose and Scope

The 718 CES is in the project validation stage of evaluating the existing Stearley Heights Elementary School (Building [B]2261), Gymnasium (B2279), and Utility building (B2285) at Kadena AB to determine how the existing spaces best suit the needs for administrative space. This project provides a visual assessment of the existing conditions and provides renovation and repair recommendations for three COAs. The COAs were developed with three paths, one COA with no significant changes, one with minimal changes, and one COA with significant changes. The visual investigations included structural, architectural, mechanical, plumbing, electrical, communication, and fire protection review and analysis.

The charrette sessions included a review of the visual investigation of the selected buildings. The functional requirements for the facility were gathered from the stakeholders and subject matter experts during the charrette work sessions and site tours and they were developed further by the Jacobs team. The goal of the charrette work sessions was to define the scope of renovation needed to bring the building up to the appropriate building code and Unified Facilities Criteria (UFC) standards.

It was confirmed by 718 CES during the planning charrette that the project will take place during Fiscal Year 2024, and that the funding stream will be centralized Operations and Maintenance (O&M). Additionally, the category code is 610-100 Administrative Office, Misc.

The renovation scope was reviewed and validated during discussions with building users by applying guidance from applicable sections in Department of the Air Force Manual (DAFMAN) 32-1084, *Facility Requirements Standards*, and UFC standards. Additional requirements were based on user input and existing conditions. Table 1-1 presents the project scope information.

Table 1-1: Convert Stearley Heights School and Gym B2261, B2279, and B2285

Installation:	Kadena AB, Japan		
Fiscal Year:	2025		
Project Name:	Convert Stearley Heights School and Gym B2261, B2279, B2285		
Project Number:	TBD		
Host Command:	Pacific Air Forces		
Requiring Command:	Pacific Air Forces		
Category Code:	CATCODE 610-100		
Scope:	**Item**	**U/M**	**Quantity**
	Convert Stearley Heights School (B2261)	SM (SF)	5,395 (58,069)
	Gymnasium (B2279)		987 (10,620)
	Utility Building (B2285)		120 (1,291)
Programmed Amount:	[redacted]		
Operational Need Date:	2027		

M = million; SF = square foot (feet); SM = square meter(s); TBD = to be determined; U/M = unit of measure

1.2.2 Description of Proposed Construction

The Stearley Heights Elementary School and gymnasium were built in 1984. A major addition to the elementary school occurred in 1996, when a one-story addition in the central portion of the facility on the south side was added for conference and classroom space. The scope for B2261 and B2285 is summarized in the following description and there are no anticipated changes for the gymnasium (B2279).

Overall, the facilities are in good condition and need basic updating of finishes and minor repairs to make them useful for the foreseeable future as an administrative facility. Excluded from the repairs are the fire protection system and heating, ventilation, and air conditioning (HVAC) system. The HVAC system will need to be completely replaced and the fire protection system does not exist at present. The HVAC components are at the end of their useable life and require replacement because of age, insufficient maintenance, and corrosion from the climatic conditions at Kadena AB. The HVAC system for B2261 requires complete replacement, including the chilled water pump, HVAC system ducts, exhaust fans, and control system. The existing facilities do not have fire sprinkler systems and fire sprinklers are required per UFC 3-600-01, Section 34-1.3; these systems will be incorporated into the recommendations. B2261 requires a new wet pipe sprinkler system, a fire pump, fire water storage tank, and fire alarm/mass notification system (MNS).

The plumbing scope includes removal of wall-hung sinks and toilets in classrooms, instantaneous water heaters for group toilet rooms, and inspection of the waste system to identify and correct any broken or blocked pipes.

The electrical scope includes adding circuits and receptacles to support additional personnel, replacing fluorescent fixtures with light-emitting diode (LED) light fixtures, and incorporating emergency lighting as necessary.

The communication system scope includes adding Cat6 outlets and distribution to support project personnel.

The structural scope includes repairing spalling concrete on the roof mechanical room, infilling the music room tiered seating to be level with the finish floor, and adding holes in the concrete parapet for scuppers.

The architectural components included for B2261 are to replace the roof drain strainers and add overflow scuppers and add an exterior access ladder to access the first-floor roof. The interior work includes a new suspended acoustical ceiling tile system throughout, new roof insulation to the underside of the structure, reconfiguring the northwest quadrant existing administrative area with an efficient administrative concept of open and private office suites, toilets, and a lactation room. Additional work includes removing the existing stove, selective demolition, selective patch-and-repair finishes, and painting the interior.

Further detail on the renovation recommendations for each building are captured in Section 2.3, Parametric Design Analysis, per the individual discipline narratives. The plan for the repairs was developed through interactive work sessions with the stakeholders to ensure that the repair items would be aligned with the planned administrative use for the facilities at Kadena AB, as well as mitigating building code and UFC deficiencies discovered during the facility walkthroughs and subsequent analysis.

1.2.3 Analysis of User Requirements

The purpose of this project is to renovate the existing Stearley Heights Elementary School and gymnasium to convert the primary use to an administrative facility to enable the 718 CES additional area for office space functions to meet the growing demand at Kadena AB. Currently, all surveyed facilities have sufficient structural, life safety, and electrical systems, with major deficiencies in the mechanical systems and fire protection systems. The mechanical system has deteriorated because of harsh climatic conditions and excessive use without consistent maintenance.

During the charrette, three COAs were developed and presented to 718 CES leadership and stakeholders for review and selection of the Preferred COA. The Preferred COA was selected after the charrette out-brief on 18 January 2024. A summary of the scope of the initial three COAs follows.

~~COA 1 includes:~~

- ~~Fire Protection:~~
  - ~~Install a new wet pipe sprinkler system throughout entire facility (fire pump and storage tank required)~~
  - ~~Install a new fire alarm/MNS throughout entire facility~~
  - ~~Add exit signage as necessary~~
- ~~Electrical:~~
  - ~~Add circuits and receptacles to building to support additional users~~
  - ~~Replace fluorescent fixtures with LED fixtures~~
  - ~~Add emergency lighting as necessary~~
- ~~Electrical changes supporting other disciplines~~
  - ~~Communications:~~
    - ~~Add Cat6 work area outlets and distribution to support additional users, up to 288 total ports~~
- ~~Mechanical:~~
  - ~~Recommission the existing mechanical systems (air and water side)~~
  - ~~Clean all existing duct systems~~
  - ~~Perform asbestos abatement from existing piping insulation~~
  - ~~Demolish the nonfunctioning boilers~~
- ~~Plumbing:~~
  - ~~Scope the waste system to identify and correct any broken or blocked pipes~~
  - ~~Implement a secondary roof drainage system~~
  - ~~Remove toilets and wall-hung sinks and lavatories within classrooms~~
  - ~~Provide instantaneous water heaters for group toilets~~
- ~~Structural:~~
  - ~~Repair concrete spalling below a blank-off louver in the upper mechanical room~~
  - ~~Add 3-inch diameter scuppers by core-drilling parapet walls at low ends of the roof areas. Apply an elastomeric protection inside the cored area.~~
- ~~Architectural:~~
  - ~~Assume 170 personnel per floor for a total occupancy of 340~~

- ~~Add a secure café on each floor~~
- ~~Assume two conference rooms per floor~~
- ~~Assume approximately 12 administrative personnel per classroom~~
- ~~Replace roof drain strainers and add overflow scupper openings~~
- ~~Add access ladder to low roof area (trapezoid)~~
- ~~Install new suspended ACT throughout~~
- ~~Plan for selective interior paint and patch finishes~~
- ~~Remove stove in B2261 first-floor kitchen~~
- ~~Incorporate lactation room~~
- ~~Remove traditional tatami mat room~~
- ~~Use existing kitchen and/or sink areas for reuse as break rooms~~

~~COA 2 includes:~~

- ~~Fire Protection:~~
  - ~~Install a new wet pipe sprinkler system throughout entire facility (fire pump and storage tank required)~~
  - ~~Install a new fire alarm/MNS throughout entire facility~~
  - ~~Add exit signage as necessary~~
- ~~Electrical:~~
  - ~~Add circuits and receptacles to building to support additional users~~
  - ~~Replace fluorescent fixtures with LED fixtures~~
  - ~~Add emergency lighting as necessary~~
  - ~~Perform electrical changes to support other disciplines~~
- ~~Communications:~~
  - ~~Add fiber optic service cabling~~
  - ~~Remove abandoned/underused communications cabling~~
  - ~~Add communications infrastructure to support maximum design capacity of users~~
  - ~~Add wireless data~~
- ~~Mechanical:~~
  - ~~Update the control system and connect it to the Base Automated Logic System~~
  - ~~Replace existing air handlers, fans, and pumps~~
- ~~Plumbing:~~
  - ~~Replace all plumbing fixtures with new low-flow type units~~
  - ~~Remove sinks built into the existing cabinets~~
- ~~Structural:~~
  - ~~Repair concrete spalling below a blank-off louver in the upper mechanical room~~
  - ~~Add 3-inch diameter scuppers by core-drilling parapet walls at low ends of the roof areas; apply elastomeric protection inside the cored area~~
- ~~Architectural:~~
  - ~~All of COA 1 items plus the following:~~
    - ~~Replace flooring throughout~~
    - ~~Demolish cabinets/sink in each classroom~~
    - ~~Replace windows to improve energy efficiency~~
    - ~~Repurpose gymnasium as a storage area~~

~~COA 3 includes:~~

- ~~Fire Protection:~~
  - ~~Install a new wet pipe sprinkler system throughout entire facility (fire pump and storage tank required)~~

        - ~~Install a new fire alarm/MNS throughout entire facility~~
        - ~~Add exit signage as necessary~~
- ~~Electrical:~~
    - ~~Add circuits to building to support additional users~~
    - ~~Replace fluorescent fixtures with LED fixtures~~
    - ~~Add emergency lighting as necessary~~
    - ~~Replace lighting controls per energy code~~
    - ~~Replace electrical breakers~~
    - ~~Perform electrical changes to support other disciplines~~
    - ~~Service transformer replacement~~
- ~~Communications:~~
    - ~~Build out telecommunications spaces~~
    - ~~Provide new backbone infrastructure~~
    - ~~Provide new communications infrastructure to support maximum design capacity of users~~
    - ~~Add wireless data~~
    - ~~Add intrusion detection and access control to utility spaces~~
    - ~~Build out common conference/video teleconference space~~
- ~~Mechanical:~~
    - ~~Perform a total upgrade to the mechanical system, including redesign of the system to variable volume type and addition of a dedicated outdoor air system (DOAS)~~
    - ~~Perform upgrades to the building thermal envelope~~
- ~~Plumbing:~~
    - ~~Redesign and replace the entire plumbing system to meet the requirements of the current codes and UFCs, including the Americans with Disabilities Act (ADA)/Architectural Barriers Act (ABA) requirements~~
- ~~Structural:~~
    - ~~Level the music room by placing a 4-inch reinforced concrete slab on styrofoam~~
    - ~~Strengthen the second-floor areas designed for 40 pounds per square foot (psf) live load to 50 psf live load~~
    - ~~Perform a detailed lateral analysis of the building for current code requirements and strengthen where needed~~
- ~~Architectural:~~
    - ~~Assume all COA 1 and COA 2 changes, as modified with the following:~~
        - ~~Increase maximum occupancy for second floor~~
        - ~~Replace all finishes~~
        - ~~Infill music room floor to match existing finish floor elevations~~
        - ~~Replace/add perimeter wall/roof insulation to match energy code requirements~~
        - ~~Demolish small toilet room and closet rooms~~
        - ~~Incorporate new dedicated electrical and communications rooms~~
        - ~~Create new partitions at the central open area to facilitate office area~~
        - ~~Add fall protection to roof~~
        - ~~Update central toilet facilities with new finishes to meet ABA standards~~

The Preferred COA was selected after discussion of the three COAs, which resulted in elements from COA 1 and COA 3 combined to form the Preferred COA. As a note, since the fire protection/sprinkler system is required for any change of use, the fire protection scope is the same for each COA. The electrical, communication, and plumbing scope items were taken from COA 1. The mechanical scope for the selected COA was taken from COA 3. Lastly, structural and architectural scope items were taken from both COA 1 and COA 3. The following provides a summary of the Preferred COA. Appendix D contains the charrette concept renovation drawings for each COA for reference.

Preferred COA:

- Fire Protection
  - Install a new wet pipe sprinkler system throughout entire facility (fire pump and storage tank required)
  - Install a new fire alarm/MNS throughout entire facility
  - Add exit signage as necessary
- Electrical:
  - Add circuits and receptacles to building to support additional users
  - Replace fluorescent fixtures with LED fixtures
  - Add emergency lighting as necessary
  - Perform electrical changes to support other disciplines
- Communications:
  - Add Cat6 work area outlets and distribution to support additional users, up to 288 total ports
- Mechanical:
  - Update the control system and connect it to the Base Automated Logic System
  - Replace existing air handlers, fans, and pumps
  - Perform a total upgrade to the mechanical system, including redesign of the system to variable volume type and addition of a DOAS
  - Perform upgrades to the building thermal envelope (roof)
- Plumbing:
  - Scope the waste system to identify and correct any broken or blocked pipes
  - Implement a secondary roof drainage system
  - Remove toilets and wall-hung sinks and lavatories within classrooms
  - Provide instantaneous water heaters for group toilets
- Structural:
  - Repair concrete spalling below a blank-off louver in the upper mechanical room
  - Add 3-inch diameter scuppers by core-drilling parapet walls at low ends of the roof areas; apply elastomeric protection inside the cored area
  - Level music room by placing a 4-inch reinforced concrete slab on styrofoam
  - Limit the live loads in the existing classrooms on the second floor to 40 psf to avoid the need for detailed floor analysis or strengthening to be fully used for 50 psf as required by UFC for office spaces
  - Consider a complete lateral analysis for wind and seismic loading, which could be grandfathered if the building is maintained at similar loading
- Architectural:
  - Maximize personnel, assume +/- 170 personnel per floor for a total occupancy of approximately 340
  - ~~Add a secure café on each floor~~
  - Add two conference rooms per floor
  - Assume approximately 12 administrative personnel per classroom
  - Replace roof drain strainers and add overflow scupper openings
  - Add access ladder to low roof area (trapezoid)
  - Install new suspended ACT throughout
  - Perform a selective interior patch and repair of finishes
  - Remove stove in B2261 first-floor kitchen
  - Incorporate a lactation room
  - Remove traditional tatami mat room
  - Reuse existing kitchen and sink areas as break rooms

- Renovate the northwest corner of the space to create an open office with private suites; add a men's and women's toilet, break room, and lactation room, and small electrical room since these areas were demolished to make room for the open office
- Add spray-applied insulation to the underside of the roof structure
- Add a communications room to the second floor
- Paint the interior

[redacted]

[redacted]

2 Primary Facilities

2.1 User Information and Space Analysis

2.1.1 Facility Descriptions

2.1.1.1 Stearley Heights Elementary School, B2261

Building 2261 is an existing 58,069 SF former elementary school referred to as Stearley Heights Elementary School. The purpose for this project is to renovate the existing school classrooms and associated support rooms into an administration office facility ready for the Air Force tenants to occupy and fulfill their mission. The interior of the facility is generally in good condition; therefore, the scope items are intended to bring value by providing efficient and cost-effective recommendations to get the Stearley Heights Elementary School up and running into an administration facility so the 718 CES can assign groups in need of short- and long-term office space.

The primary drivers used in developing the Preferred COA are as follows:

- Maximize the number of personnel that can occupy the facility
    - The Preferred COA shows 177 personnel on the first floor and 170 personnel on the second floor, for a total occupancy of 347 (note that expanding the parking for the building is not included in the scope and it will limit the number of people who can be assigned)
- ~~Designate a room on each floor that can be used as a secure café~~
- Designate two conference rooms per floor
- Target +/- 12 personnel for each existing classroom for admin conversion
- Use the existing kitchen/kitchenette areas for the proposed break rooms to capitalize on the existing plumbing and infrastructure already in place
- Assume normal operating hours are 06:30 to 18:00
- Assume reuse of existing furniture provided by government; no equipment is expected

2.1.1.2 Stearley Heights Gymnasium/Cafeteria, B2279

The existing gymnasium and cafeteria building is 10,620 SF and will remain as a gymnasium for the community. The existing seating for the cafeteria is no longer in use and the kitchen area is currently being used as storage, so it will remain as such. No scope items for B2279 are included in this effort.

2.1.1.3 Utility Building, B2285

The utility building is 1,291 SF and contains the chiller water pumps and heat exchangers for the school. These will be replaced as part of the HVAC recommendations for this project.

2.1.2 Renovation Sketches

The graphics presented on the following pages depict conceptual facility plans based on the information-gathering, stakeholder input, and analytical processes of the charrette and post-charrette planning activities.

The recommended renovation scope items are primarily interior; therefore, conceptual floor plans are presented, and the existing site will remain as is currently configured. The following diagrams represents the Preferred COA.

The building configurations depicted in this concept are based on stakeholder input, flexibility of user group functions, adaptability for future missions, and total building utilization. Building concepts identify the planning analyses for each of the functional areas of each facility and establish the foundation for further development during the design phase. Jacobs assumes that the final design concept will comply with relevant codes and criteria, as well as other applicable guidelines.

Figure 2-1: Preferred COA – First Floor

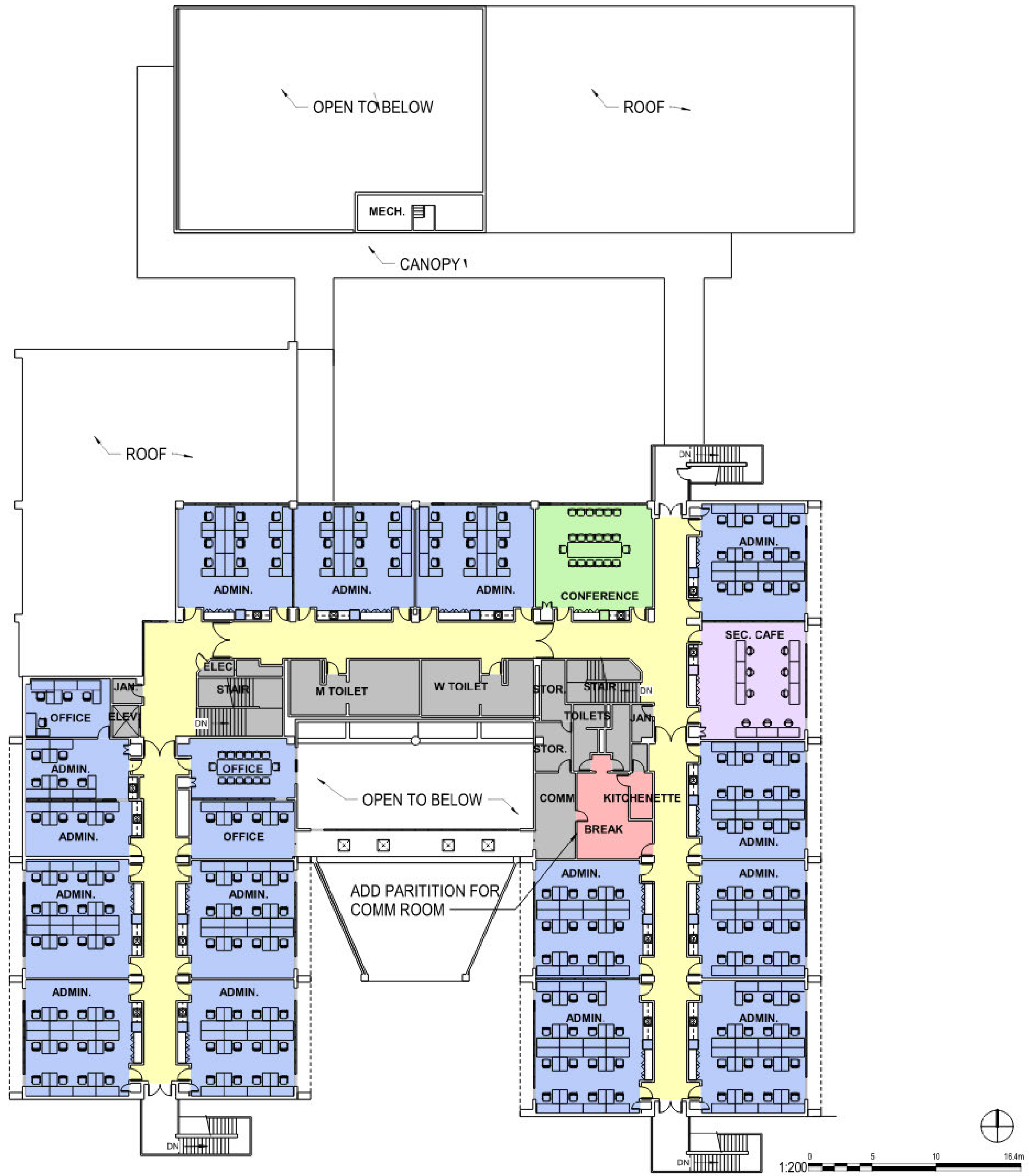

Figure 2-2: Preferred COA – Second Floor

2.2 Parametric Design Analysis

This section provides a conceptual-level narrative that describes identified requirements and notional design features or assumptions made for cost-estimating purposes. The DA is responsible for the final design solution to meet the project requirements.

2.2.1 Structural

2.2.1.1 Stearley Heights Elementary School, B2261

The existing facility is a two-story building with cast-in-place load-bearing concrete walls and cast-in-place concrete framing for the second floor and roof. The ground floor consists of cast-in-place reinforced slab-on-grade.

The rigid perimeter walls comprise the lateral force resisting system in all directions. The concrete rigid frames structural system transfers lateral loads to the foundations.

2.2.1.1.1 Substructure General Requirements

The foundation system for this building consists of shallow spread footings for the interior concrete columns and continuous footings for the concrete walls. No foundation movement was observed. The elevated slabs and the concrete slab-on-grade appear to be in good condition.

2.2.1.1.2 Superstructure and Roof General Requirements

The second and roof floor framing consists of cast-in-place reinforced concrete slab spanning between concrete columns and concrete walls. The existing facility comprises a two-story cast-in-place concrete framed structure with interior cast-in-place concrete columns and walls and perimeter cast-in-place concrete walls and columns to support the elevated cast-in-place concrete slabs at the second floor and roof. The lateral force resisting system consists of cast-in-place concrete shear walls in both directions.

The second floor and roof system consist of cast-in-place concrete slabs supported by cast-in-place concrete beams, which frame between cast-in-place concrete walls and columns. The rigid perimeter walls and columns comprise the lateral force resisting system in all directions. The concrete rigid frames structural system transfers lateral loads to the foundation.

2.2.1.1.3 Structural Modifications and Repairs

This existing facility is in a high seismic area with Seismic Design Category E designation. It can be subjected to winds up to 283 miles per hour for Risk Category II, based on current UFC requirements. Based on seismic and wind forces, lateral loads are resisted by the concrete wall rigid frames. Because this facility was designed and built between 1984 and 1986 to function as a school, and the proposed conversion/study is to change the occupancy to function as an administrative space, a more in-depth structural analysis is recommended to verify if any structural upgrade is required.

The initial occupancy of the second floor was to be used for classrooms, which were designed for 1.9 kilopascals (kPa) (40 psf) uniform live load. The proposed use as offices requires a design load of 2.4 kPa (50 psf) plus partition loads. It is recommended that an evaluation of the second-floor structural system be performed to verify if the existing system is capable of supporting the higher loads. If it is determined that it cannot support the higher loads, a strengthening program will be recommended.

The existing music room has two concrete stairs to access a lower floor and step floors up to ground floor. This area will be leveled by infilling the low space and pouring a concrete slab at the same elevation of the ground floor.

For the upper mechanical room that is accessed from the roof, it was observed that the concrete wall below a blank-off louver is experiencing some concrete spalling and rusted rebar caused by water leakage. The wall area of approximately 1.2 square meters needs to be repaired.

When it rains, it was observed that large quantities of water accumulate at the lower portion of the sloped roof. It is recommended that some 3-inch (75 millimeter [mm]) diameter holes be core drilled throughout the parapet walls to allow water to drain. The exposed cored holes should be protected with an elastomeric membrane of the same color as the building to protect the concrete and any reinforcing steel that is cut.

2.2.1.1.4 Seismic Design Requirements

The project site is in a high seismic area with a Seismic Design Category E designation. Based on seismic forces, lateral loads should be resisted by the concrete wall rigid frames. All nonstructural components should require seismic bracing. A seismic evaluation in accordance with American Society of Civil Engineers (ASCE) Standard 41 should be performed for the entire building during the design/evaluation phase.

2.2.1.1.5 Structural Antiterrorism Requirements

Protective measures as specified by the current Department of Defense (DoD) Minimum Antiterrorism Standards for Buildings (UFC 4-010-01) are required for this facility. However, since the conversion cost is estimated to be less than 50% of the estimated replacement cost (ERC), these protective measures will not apply for this effort. Because the structure is less than three stories in height, it is not required to be evaluated to resist progressive collapse in accordance with UFC 4-023-03.

2.2.1.1.6 Special Structural Requirements

Per UFC 1-200-01, Chapter 4, Kadena AB has an Environmental Severity Classification (ESC) of C5. The corrosive environment in Okinawa, Japan, must be considered to protect the rebar and steel from rusting.

2.2.1.2 Stearley Heights Gymnasium/Cafeteria, B2279

The existing facility consists of a single high-bay building with cast-in-place load-bearing concrete walls and cast-in-place concrete framing for the roof. The ground floor consists of cast-in-place reinforced slab-on-grade.

The rigid perimeter walls comprise the lateral force resisting system in all directions. The concrete rigid frames structural system transfers lateral loads to the foundations.

2.2.1.2.1 Substructure General Requirements

The foundation system consists of pile and pile caps to support the upper framing, but no foundation movement was observed. The slabs are concrete slab-on-grade, which appear to be in good condition.

2.2.1.2.2 Superstructure and Roof General Requirements

The roof framing consists of cast-in-place reinforced concrete slab spanning between concrete walls. The lateral force resisting system consists of cast-in-place concrete shear walls in both directions.

2.2.1.2.3 Structural Modifications and Repairs

It is anticipated that no structural modifications will be required for this facility.

2.2.1.2.4 Seismic Design Requirements

This existing facility is in a high seismic area with a Seismic Design Category E designation. Based on seismic forces, lateral loads should be resisted by the concrete wall rigid frames. All nonstructural components should require seismic bracing.

2.2.1.2.5 Structural Antiterrorism Requirement

Protective measures as specified by the current DoD Minimum Antiterrorism Standards for Buildings (UFC 4-010-01) are required for this facility. However, protective measures will not apply to this effort. Because the structure is less than three stories in height, it is not required to be evaluated to resist progressive collapse in accordance with UFC 4-023-03.

2.2.1.2.6 Special Structural Requirements

Per UFC 1-200-01, Chapter 4, Kadena AB has an ESC of C5. The corrosive environment in Okinawa, Japan, must be considered to protect the rebar and any miscellaneous steel elements from rusting.

2.2.1.3 Utility Building, B2285

The existing facility consists of a single-story building with cast-in-place load-bearing concrete frame and infill precast walls and steel framing for the roof. The ground floor consists of cast-in-place reinforced slab-on-grade.

The rigid perimeter concrete framing comprises the lateral force resisting system in all directions. The concrete rigid frames structural system transfers lateral loads to the foundations.

2.2.1.3.1 Substructure General Requirements

The foundation system consists of shallow continuous footing to support the concrete framing. No foundation movement was observed. The slabs are concrete slab-on-grade, which appears to be in good condition.

2.2.1.3.2 Superstructure and Roof General Requirements

The roof framing consists of steel framing to support the metal deck roofing.

2.2.1.3.3 Structural Modifications and Repairs

No apparent repair needs were observed at this facility.

2.2.1.3.4 Seismic Design Requirements

This existing facility is in a high seismic area with a Seismic Design Category E designation. Based on seismic forces, lateral loads are resisted by the rigid concrete wall frames. All nonstructural components should require seismic bracing. A seismic evaluation in accordance with ASCE 41 should be performed for the entire building during the design/evaluation phase.

2.2.1.3.5 Structural Antiterrorism Requirements

Protective measures as specified by the current DoD Minimum Antiterrorism Standards for Buildings (UFC 4-010-01) are not required for this facility since the building is not occupied.

2.2.1.3.6 Special Structural Requirements

Per UFC 1-200-01, Chapter 4, Kadena AB has an ESC of C5. The corrosive environment in Okinawa, Japan, must be considered to protect the rebar and any miscellaneous steel elements from rusting.

2.2.2 Architectural

The project and Department of Defense (DD) Form 1391 scope consists of renovating and sustaining existing facilities; therefore, the existing architectural style, massing, and fenestration/window openings should be maintained.

The facility renovation should be designed to be noncombustible, have a 50-year useful life, and be low maintenance with durable material choices. The climate is hot and humid; therefore, the facilities should be resistant to tropical conditions, including high winds, high humidity, high temperatures, corrosive salt-laden air, and earthquakes.

Materials should be selected for ease and speed of construction and should resist fire, winds, and seismic conditions.

Dead-level roofs and rooftop-mounted equipment are not allowed. The minimum roof slope for low-slope roofs should comply with UFC 3-110-03. The existing B2261 roof does appear to comply with minimum slope requirements based on the as-builts and onsite visual observation.

A satisfactory building envelope requires strength, durability, and weather resistance. The vapor retarder, insulation, and vapor dissipation systems should be properly balanced. Since this project is a renovation, the air barrier should follow the best practice guidance in Appendix A of UFC 3-101-01 for modifying an existing building to any area that is disturbed or repaired for the building envelope.

Any field-applied paints should be flexible enough to expand and contract with the substrate. The need for paint should be avoided by using materials that are in their natural color, are prefinished, or are integrally colored. The exterior color scheme should comply with the *Kadena Air Base Design Guide* and match the existing facilities color scheme, which also should be used to guide the design of each renovation.

The Preferred COA concept plans, the proposed demolition plans, and the proposed roof plan are included in Appendix D.

2.2.2.1 Exterior Enclosure and Roofing

The following architectural elements represent the recommended approach for the renovated materials/systems and the quality of each exterior building component.

2.2.2.1.1 Exterior Walls

The existing walls are cast-in-place concrete with a spray acrylic coating and are in good condition. In general, no new work is anticipated for the exterior walls except where new penetrations or attachments are required. This includes new overflow drains, a roof access ladder, and any required mechanical penetrations. New penetration should be patched, repaired, and painted with a silicon-based elastomeric coating with antimicrobial properties and painted to match the adjacent surface.

2.2.2.1.2 Roof Assembly

The roof structure is cast-in-place concrete with a spray-applied roofing membrane. The roofing membrane is in good condition and no additional work is recommended. If any roof penetrations should be required during the design and construction, the roofing should be patched and repaired with a single-component, elastomeric, air-dry silicone coating with antimicrobial properties directly applied to the roof deck with a solar reflective top coating. Any flashings and copings should be prefabricated and factory finished. The existing roof does not appear to have insulation per the as-built drawings. The insulation is shown at the interior ACT level; therefore, it is recommended to install a spray-applied roofing insulation to the underside of the roof structure at approximately an RSI 6.7 (R-38) to provide a thermal envelope to meet the energy-saving requirements of American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) Standard 90.1.

2.2.2.1.3 Roof Drainage

The existing roof drains appear to be located and sized adequately, but the roof drain strainers are in corroded condition and are preventing the roof drainage from occurring efficiently. Some locations have backed up during heavy rains and caused leaks or overflow. To correct this problem, two actions are recommended. Firstly, each roof drain strainer should be replaced. Secondly, there are no overflow drains on the existing roof; an overflow scupper is recommended to be incorporated by drilling 3-inch (75 mm) diameter holes into the concrete parapet wall just above the roof drains. This will allow overflow water to spill over the side in case of a clogged drain and also will alert personnel that there is a problem on the roof that needs maintenance.

2.2.2.1.4 Doors

No work is anticipated. Any unforeseen replacement of personnel doors and frames should be aluminum and have an anodized finish. Doors should be insulated and weather stripped with integral thermal break.

2.2.2.1.5 Keying Hardware

No changes are anticipated, but any keying updates/changes should be coordinated with 18 CES.

2.2.2.1.6 Windows

The existing windows are aluminum frame with a single-pane glazing component and will remain in place based on the limited scope of this project. One benefit to the existing building configuration is that the windows are shaded by an extensive concrete block shade structure on most of the inhabited spaces, resulting in little direct sun exposure. If any windows are damaged during construction, the window frames should be aluminum with impact-resistant, insulated, low-E, laminated glass with reflective tint. Glazing thickness should be designed to meet typhoon resistance requirements.

2.2.2.1.7 Louvers

No work is anticipated. Any louver required to be added during the design should be stainless steel or aluminum material and should be typhoon and insect resistant. Louvers should provide supporting performance data in accordance with the Air Movement and Control Association (AMCA) Standards 500 and 511, and louvers should bear AMCA certified rating seals for air performance and water penetration.

2.2.2.1.8 Metals

A roof access ladder is recommended to gain access to the one-story roof above the library (1996 addition area). All exterior metal elements should be aluminum or stainless steel grade 316L for superior corrosion resistance, with special attention to providing a means of separation for any dissimilar metals to avoid galvanic action and premature corrosion.

2.2.2.1.9 Exterior Scope Summary

The following subsections present the recommended exterior scope summary for each facility.

2.2.2.1.9.1 Stearley Heights Elementary School, B2261

The exterior architectural elements are in good condition and work will be required only where the exterior components are adjacent or connected to the following scope items:

- Replace the existing roof drain strainers
- Incorporate scuppers (3-inch/75 mm diameter holes) in the parapet at each roof drain and patch/paint with elastomeric coating
- Install a roof access ladder to the one-story trapezoid roof structure (south central side of building); clean, patch, repair, and repaint concrete walls with elastomeric coating at any new penetrations required

2.2.2.1.9.2 Stearley Heights Gymnasium/Cafeteria, B2279

There are no current renovations anticipated for the gymnasium/cafeteria. It will continue to house the gymnasium activities and ground-level or low-shelving storage will continue in the cafeteria area.

2.2.2.1.9.3 Utility Building, B2285

The exterior architecture scope for the utility building will be limited to patch, paint, and repair where any new penetrations are required for the new HVAC system.

2.2.2.2 Interior Construction, Stairs, and Finishes

The primary interior scope items for the architectural components are as follows:

- Replace the ceilings based on the new fire sprinkler system required to be installed
- Add roof insulation to the underside of the roof structure
- Reconfigure the northwest administration quadrant
- Paint/patch/repair any finishes for the demolition/replacement of any scope recommendations required for all disciplines

The existing northwest quadrant configuration is not an efficient layout and contains multiple smaller rooms, three toilet compartments, and three millwork/kitchenette areas. One request from 718 CES is to provide some private offices for flexibility for tenant configurations. The proposed recommendation provides five private offices along with an open workstation area with 24 workstations. This provides more density and flexibility with the adjacent private offices. Additionally, the reconfiguration allows for a break area, consolidated men’s and women’s toilets, and a lactation room that is not currently present at the facility and is required for DoD facilities. The office area and lactation room finishes should be vinyl composition tile flooring, painted gypsum board walls, and a suspended acoustical tile ceiling. The toilet rooms should have ceramic tile flooring and wainscotting with moisture-resistant painted gypsum board above. The ceiling should be painted moisture-resistant gypsum board.

The existing classrooms are well suited to be converted into administration space and each can hold approximately 12 workstations. The existing built-in cabinets and sinks in each classroom will be left as is to be used as storage for each administration suite. However, there are wall-hung lavatories outside of each toilet compartment in the classrooms that will be removed to create an efficient, open space for the incoming furniture and administration workstations. The existing single kid-sized toilets also will be removed and capped to allow those rooms to be used as storage.

~~The secure café rooms are purposefully stacked on the first and second floors in case there is a need to provide separation from other suites. There is currently an ongoing project to convert the second-floor room designated in the concept plan to a secure café. Additionally, the first-floor secure café is designated at that location, but there are no security requirements required at this time. For this effort, black-out curtains for those two rooms will be included, but no additional effort will be made for the walls/windows/doors, which will have to be addressed and analyzed by the System Safety Manager for the facility to confirm if any additional secure café requirements are required for the future tenants.~~

~~Additional recommendations include:~~

- ~~Remove the Host Nation tatami mat room on the first floor to provide additional open area to configure new office layouts efficiently~~
- ~~Remove the stove in the kitchen on the first floor to remove potential code violation issues~~

- ~~Add a communications room on the second floor since very little room is available for the communications infrastructure~~
- ~~Paint the interior to provide a fresh and updated appearance with a cost-effective solution~~

~~The flooring throughout the facility is typically vinyl composition tile and is in adequate shape.~~

2.2.2.2.1 Interior Scope Summary

The following subsections present the recommended interior scope summary for each facility.

2.2.2.2.1.1 Stearley Heights Elementary School, B2261

- Install a new suspended acoustical tile ceiling throughout
- Paint, patch, and repair finishes as needed for damaged areas or where fixtures/walls have been demolished
- Remove the stove from the first-floor kitchen
- Incorporate a lactation room
- Remove the traditional tatami mat room and repair finishes for administration space
- Revise the northwest quadrant of school administrative space to create an open and private office suite. Add a men's and women's toilet, break room, and lactation room, and a small electrical room to replace rooms that are to be demolished
- Add spray-applied insulation to the underside of the roof structure
- Remove wall-hung lavatories in classrooms
- Add a communications room to the second floor
- Paint interior walls
- ~~Add black-out curtains to the secure café rooms~~

2.2.2.2.1.2 Stearley Heights Gymnasium/Cafeteria, B2279

There are no current renovations anticipated for the gymnasium/cafeteria. It will continue to house the gymnasium activities and ground-level or low-shelving storage will continue in the cafeteria area.

2.2.2.2.1.3 Utility Building, B2285

- The interior architecture scope for the utility building will be limited to patch, paint, and repair where any new penetrations are required for the new HVAC system.

2.2.2.3 Accessibility

Any renovated areas of the facilities will be handicapped accessible for civilian employees, visitors, and Warriors-in-Transition that may be assigned in accordance with the ABA standard for DoD facilities.

2.2.2.4 Secure Compartmented Information Facility

Not applicable.

2.2.2.5 State Historical Preservation Office Review

Not applicable.

2.2.2.6 Installation Facility Standards

The project should comply with the *Kadena AB Architectural Compatibility Guide* and the *Kadena AB Design Guide*.

2.2.3 Conveying

There is an existing elevator that will remain in B2261.

2.2.4 Plumbing

2.2.4.1 Stearley Heights Elementary School, B2261

The plumbing system consists of water supply and waste and vent systems for several group and individual latrines. The plumbing systems appear to be original to the building, with a limited number of fixtures having been converted to be ADA/ABA compliant. However, the ADA/ABA fixtures are not fully compliant with the current accessibility requirements (missing protective insulation).

The storm drainage system is via collection boxes at the building perimeter. The building lacks a secondary storm drainage system.

Hot water generation was designed to be via a water-to-water heat exchanger with mechanical boilers and a heat recovery chiller as the main heating sources. The mechanical boilers and the heat recovery chiller, and therefore the domestic hot water system, are currently nonfunctional.

The sanitary waste system must be scoped using cameras to establish its integrity and identify any potential problems with blockage of pipes or lack of sufficient slope in the waste piping system. Any broken or blocked pipes identified in this process must be repaired.

The primary roof drain system should be modified with addition of scuppers to act as a secondary roof drainage method.

The existing wall-hung sinks and lavatories located in the individual classrooms are to be demolished.

The existing nonfunctional domestic hot water system must be abandoned in favor of new water heaters serving the remaining group latrines. The water heaters can be small heaters dedicated to each bathroom group or point-of-use instantaneous water heaters.

2.2.4.2 Stearley Heights Gymnasium/Cafeteria, B2279

The plumbing system consists of water supply and waste and vent systems for several group latrines and a currently nonfunctioning kitchen. The plumbing systems appear to be original to the building, with some fixtures having been converted to be ADA/ABA compliant. However, the ADA/ABA fixtures are not fully compliant with the current accessibility requirements (missing protective insulation).

Hot water generation was designed to be via a water-to-water heat exchanger with mechanical boilers and a heat recovery chiller as the main heating sources. The mechanical boilers and the heat recovery chiller, and therefore the domestic hot water system, are currently nonfunctional.

2.2.4.3 Utility Building, B2285

The plumbing systems appear original to the building, with major components such as the heat exchanger having been replaced at a later time. The water supply system serves the heat exchanger and provides makeup water to mechanical equipment and some miscellaneous hose bibbs.

2.2.5 Heating, Ventilation, and Air Conditioning

2.2.5.1 Stearley Heights Elementary School, B2261

The main cooling source for the building is chilled water produced by three chillers located at the utility building yard. Two of the chillers are of conventional design and the third is a heat recovery chiller that was intended to produce hot water to supplement domestic hot water needs. Each chiller was originally designed for a nominal 85-ton capacity.

The heat recovery chiller has since been removed and is no longer present. The two remaining chillers have been recently replaced and appear to be in good condition. The new chillers have a nominal capacity of 110 tons each.

The main heating source for the building consisted of the hot water system fed by two hot water boilers of 850,000 British thermal units per hour capacity each. These boilers are no longer functional and are abandoned.

Hydronic water is supplied to four air handling units located on the roof. Each air handler consists of a cooling coil, a heating coil, and a constant-volume supply fan. Conditioned air from these air handlers is distributed throughout the building via a system of low-pressure ductwork. The current design does not allow for individual zone temperature control. Additionally, the outside air delivery system is not compliant with today's UFC requirements.

The entire building's thermal envelope needs to be upgraded to meet current ASHRAE and UFC standards for sustainable construction.

The entire HVAC system for the building, including the duct system, needs to be replaced with a new system compliant with the latest codes and UFC requirements. Based on the high-humidity climate of this location, use of a DOAS supplying preconditioned, neutral air directly to the occupied spaces is recommended. The use of a DOAS is mandatory for outside air flows greater than 750 cubic feet per minute (cfm) per UFC requirements.

The new HVAC system is recommended to be a variable volume system with new air handlers located in the current mechanical rooms on the roof of the building. New HVAC system shell be chilled water system to cover entire facility with 208 volt equipment and individual VAV control for every open office area. Individual spaces must be optimally grouped into spaces with thermally equivalent profiles to afford effective controllability of all spaces.

The building's control system should be updated to accommodate the modified and added equipment and be integrated with the existing campus Control Logic System. Provide control system such as BACnet /MSTP or MODBUS instead of BACnet /IP to connect to an existing DDC system using BACnet/MSTP or MODBUS.

Building exhaust fans need to be replaced with new fans. Low leakage dampers need to be added to all fresh air and exhaust air systems to close upon activation of the MNS.

2.2.5.2 Stearley Heights Gymnasium/Cafeteria, B2279

The main cooling source for the building is chilled water produced by three chillers located at the utility building yard, as mentioned previously for B2261.

The main heating source of the building consisted of the hot water system fed by two hot water boilers in the utility building, as described previously for B2261. These boilers are no longer functional and are abandoned.

Hydronic water is supplied to two air handling units located in the upper mezzanine area. Each air handler consists of a cooling coil, a heating coil, and a constant-volume supply fan. Conditioned air from these air handlers is distributed throughout the building via a system of low-pressure ductwork. The air handling systems are currently not functioning. The gymnasium is currently cooled by several ductless split systems. These systems do not comply with the current code requirements for outside air supply.

The existing air handling units in this building are to be replaced with new units with capacities equal to the existing units.

2.2.5.3 Utility Building, B2285

The mechanical system for the utility building is limited to a thermostatically controlled ventilation fan.

The utility building houses the chilled water pumps serving B2261. These pumps are to be replaced as part of this renovation.

2.2.5.4 Mechanical Antiterrorism Requirements

2.2.5.4.1 Stearley Heights Elementary School, B2261

The existing building has no provision for shutting off outside, exhaust, and relief air streams with low leak dampers as required by Standard 18 of UFC 4-010-01. The addition of these dampers is recommended.

2.2.5.4.2 Stearley Heights Gymnasium/Cafeteria, B2279

The existing building has no provision for shutting off outside, exhaust, and relief air streams with low leak dampers as required by Standard 18 of UFC 4-010-01. The modifications to this building's mechanical systems are deemed outside the scope of this effort.

2.2.5.4.3 Utility Building, B2285

The existing building has no provision for shutting off outside, exhaust, and relief air streams with low leak dampers as required by Standard 18 of UFC 4-010-01. The modifications to this building's mechanical systems are deemed outside the scope of this effort.

2.2.6 Fire Protection and Life Safety

All facilities have been evaluated for their existing conditions pertaining to fire suppression, fire alarm, and life safety. The following subsections summarize the existing conditions of systems and their requirements.

The scope of changes is defined by UFC 3-600-01, Chapter 34 (Existing Facilities); National Fire Protection Association (NFPA), Standard 101, Chapter 43 (Building Rehabilitation); and the International Building Code (IBC). Because the facility is undergoing a change in use, UFC 3-600-01, Chapter 34-1.3.1, requires the area of change to comply with the requirements for new construction. The entire Group E (IBC occupancy)/educational (NFPA 101) facility is being repurposed as a Group B (IBC)/Business (NFPA 101) facility and will comply with the requirements for new construction as defined by UFC 3-600-01, NFPA 101, and IBC. The NFPA 101 Building Rehabilitation classification is Change in Use per NFPA 101, Chapter 43.7.

Stearley Heights gymnasium and utility building will not undergo extensive changes. Work will be performed in these buildings such that the existing building and life safety systems are not made less compliant than systems previously installed.

2.2.6.1 Stearley Heights Elementary School, B2261

2.2.6.1.1 Fire Suppression

The existing Stearley Heights Elementary School is a non-sprinklered facility. Because the facility is multiple stories, UFC 3-600-01, Section 9-7.2.1.2, requires complete automatic sprinkler protection.

An automatic wet pipe fire sprinkler system should be designed and installed throughout B2261 in accordance with UFC 3-600-01 and NFPA 13. A riser check valve assembly should be provided for each story. The fire sprinkler system should be supervised by the building fire alarm/MNS.

Fire sprinkler piping should be black steel, minimum Schedule 10 for all piping 2.5 inches and greater in diameter and minimum Schedule 40 for all piping 2 inches and smaller in diameter. Seismic bracing should be provided on all fire sprinkler piping per NFPA 13. A wall-mounted fire department connection should be installed to serve all the new wet pipe sprinkler risers.

Areas of B2261 should be individually classified by hazard and protected in accordance with NFPA 13 and UFC 3-600-01. Light hazard spaces, including offices, corridors, restrooms, break rooms, and similar spaces, should be hydraulically calculated to provide 0.1 gallon per minute (gpm)/SF over the most remote 1,500 SF. Sprinklers in light hazard spaces should be k-factor 5.6, quick-response, standard spray upright or pendent at spacing not exceeding 225 SF per sprinkler. Ordinary hazard spaces, including building utility equipment spaces, general storage, janitor's closets, and similar spaces, should be hydraulically calculated to provide 0.2 gpm/SF over the most remote 2,500 SF. Sprinklers in ordinary hazard spaces should be k-factor 8.0, quick-response, standard spray upright or pendent at spacing not exceeding 130 SF per sprinkler.

The maximum estimated fire sprinkler demand is approximately 850 gpm at 50 pounds per square inch (psi). The awarded qualified fire protection engineer (QFPE) should perform detailed hydraulic calculations to verify the adequacy of the existing water supply.

2.2.6.1.2 Fire Alarm/Mass Notification System

The facility is equipped with an active conventional fire alarm system (Nittan). The fire alarm system is designed to a combination of U.S. and Japanese standards. The fire alarm system monitors manual stations, heat detectors, and smoke detectors for activation or occupant notification. The fire alarm system also has smoke control functionality for the classroom/conference room addition. The fire alarm system has an external Monaco transmitter for signal transmission to the Kadena AB Fire Department.

UFC 3-600-01, Chapter 9-18.4.1, requires fire alarm systems to be addressable and Chapter 9-18.1.2 requires fire alarm systems to be combined with MNSs where both are required. Based on these requirements, the existing conventional fire alarm system should be demolished and replaced with a new combined addressable fire alarm/MNS to serve the entire facility. The existing Monaco transmitter is permitted to remain in service and be reconnected to the new fire alarm/MNS.

The fire alarm/MNS should be designed and installed in accordance with the requirements of NFPA 72, UFC 3-600-01, and UFC 4-021-01. The control panel should be in a conditioned space readily accessible by the Kadena AB Fire Department.

The fire alarm/MNS control panel should monitor the following alarm signals for the facility: smoke detection above fire alarm panels and power supplies, waterflow switches on wet pipe sprinkler risers, manual pull stations within 5 feet of all exterior doors, and fire pump running. Activation of alarm signals should automatically transmit a signal to the Kadena AB Fire Department and initiate MNS messages throughout the facility.

The fire alarm/MNS control panel should monitor the following supervisory signals for the facility: duct smoke detectors in air handling units greater than 2,000 cfm per NFPA 90A, fire water storage tank level and temperature, activation of local operating console (LOC)/MNS, and fire pump supervisory. Activation of supervisory signals should automatically transmit a signal to the Kadena AB Fire Department, but not initiate MNS messages.

The fire alarm/MNS control panel should monitor the following trouble signals for the facility: fire alarm power loss, fire alarm battery loss, fire alarm circuit loss, fire alarm ground fault, and fire pump trouble. Activation of trouble signals should automatically transmit a signal to the Kadena AB Fire Department, but not initiate MNS messages.

The fire alarm/MNS should connect to a notification appliance network, including white speakers throughout the building marked "ALERT" in red lettering, to achieve a minimum intelligibility score (CIS) of 0.80 per UFC 4-021-01. Based on the facility construction, occupancy, finishes, and size, there are no spaces in the facility that should require CIS reductions per UFC 4-021-01.

The fire alarm/MNS also should include supplementary MNS features, including textual notification appliances above all substantial means of egress and LOCs to allow building occupants to transmit live voice messages. LOCs should be located such that the travel distance does not exceed 200 feet to reach one from any point within the building. Live voice messages from the LOCs should override any automatic or prerecorded MNS messaging.

2.2.6.2 Stearley Heights Gymnasium/Cafeteria, B2279

2.2.6.2.1 Fire Suppression

The existing Stearley Heights gymnasium is a non-sprinklered facility. The extent of work in the gymnasium is classified as a renovation and does not require the addition of a sprinkler system to the space.

2.2.6.2.2 Fire Alarm/Mass Notification System

The existing Stearley Heights gymnasium is protected by the same conventional fire alarm system as the existing Stearley Heights Elementary School. In accordance with NFPA 101, Chapter 43.4.1.4, work must not make the building less conforming than previous approved installations. The demolition of the conventional fire alarm system in Stearley Heights Elementary School requires the new fire alarm/MNS to be extended into the gymnasium.

2.2.6.3 Utility Building, B2285

2.2.6.3.1 Fire Suppression

The existing utility building is a non-sprinklered facility. The extent of work in the utility building is classified as a renovation and does not require the addition of a sprinkler system to the space.

2.2.6.3.2 Fire Alarm/Mass Notification System

The existing utility building is not equipped with a fire alarm system. The extent of work in the utility building is classified as a renovation and does not require the addition of a fire alarm system to the space.

2.2.6.4 Installation Water Supply

A fire hydrant flow test was performed in accordance with UFC 3-600-01 and NFPA 291 to verify the adequacy of the existing water distribution system to the project site. The flow test was performed using two hydrants along the east side of B2261. The dry barrel fire hydrant nearest the main entry vestibule was selected as the test hydrant, while the wet barrel fire hydrant at the southeast corner of the facility was selected as the flow hydrant.

The fire hydrant flow test indicated a static pressure of 30 psi and a residual pressure of 19 psi while flowing 628 gpm. The test results indicate an available flow of 600 gpm at 20 psi. The test was conducted on 17 January 2024 at 1400 hours. The facility fire flow is 1,375 gpm at 20 psi per UFC 3-600-01 and NFPA 1 and the facility fire water demand is an estimated 850 gpm at 50 psi per UFC 3-600-01 and NFPA 13.

The fire hydrant flow test results indicate that the existing water distribution system is not adequate to support either the manual fire flow or automatic fire water demands required at the facility. A fire water storage tank and fire pump are required to supply the site fire water demands.

The fire water storage tank will be designed and installed in accordance with NFPA 22 and UFC 3-600-01. The tank will be sized to hold a minimum of 120% of the required capacity per UFC 3-600-01. The maximum capacity served is the fire flow, which requires a stored capacity of 198,000 gallons (1,375 gpm * 120 minutes * 1.2).

The fire pump will be designed and installed in accordance with NFPA 20 and UFC 3-600-01. The pump will be sized to support the fire flow water demand (1,375 gpm) and fire sprinkler pressure demand (50 psi). A fire pump rated for 60 psi at 1,000 gpm is planned for the facilities.

The test area was undergoing construction activities at the time of the test and the Base escort could not verify that all waterlines and distribution sources were active. During the design phase, the awarded QFPE will perform another fire hydrant flow test to verify the adequacy of the water distribution system for fire water demands.

2.2.6.5 Life Safety Code, NFPA 101 and Building Code Analysis

UFC 3-600-01, Section 3-1, requires that the occupancy classifications for the buildings be determined in accordance with the IBC and NFPA 101 (Life Safety Code [LSC]). In accordance with UFC 3-600-01, the IBC should be used to determine the occupancy classification as it relates to allowable construction type, building height, building area, building separation distance, occupancy separation, and associated requirements. In accordance with UFC 3-600-01, Section 3-3.1.1, the occupancy classification should be determined in accordance with the LSC as it relates to fire/smoke resistance rating of interior non-load-bearing partitions (other than occupancy separation), means of egress, interior finish, features of fire protection (including vertical openings), and associated requirements.

2.2.6.5.1 Building Code Analysis (IBC)

The work in B2261 will result in a single-use Group B facility with incidental Groups A-3 and S-1 in accordance with IBC Section 509. All building code features will conform to the requirements for a new Group B facility. The existing building is two-story, Type IIB construction, and 58,069 SF.

The existing height, area, and number of stories complies with the tabular requirements for a Group B facility of Type IIB construction per IBC Tables 504.3, 504.4, and 506.2. No modifications, considerations for frontage, or considerations for unlimited area buildings are required.

The most restrictive fire separation distance is 17.5 feet between B2261 and B2279. IBC Table 705.5 permits 0-hour exterior walls between occupancy Groups E and B for fire separation distances greater than 10 feet. No modifications are required for fire separation distances.

There are no required occupancy separations in a single-use Group B building. No new fire barriers are required for this scope of work.

2.2.6.5.2 Life Safety Code Analysis

The work in B2261 will result in a single-use Business facility in accordance with NFPA 101, Chapter 38. All LSC features will conform to the requirements for a new Business facility. The existing building is two-story, Type IIB construction, and 58,069 SF.

Means of egress should be in accordance with NFPA 101, Chapter 7 and Chapter 38. The first floor has five existing sets of 32-inch double doors, which can accommodate 320 occupants each for a total available egress capacity of 2,160 occupants. The estimated occupant load factor for the first floor is 225. No additional exits are required from the first floor.

The second floor has three existing 51-inch exterior stairs, which can accommodate 170 occupants each for a total available egress capacity of 510 occupants. The estimated occupant load for the second floor is 219. No additional exits are required from the second floor.

The exterior stairs are permitted to be unenclosed with no fire resistance rating by NFPA 101, Chapter 7.2.2.6.3.1(2). The egress capacity of the second floor does not consider the two Convenience Stairs, which are constructed in accordance with NFPA 101, Chapter 8.6.9.2, and permitted by NFPA 101, Chapter 38.3.1.1(2).

The existing arrangement of means of egress is compliant with the requirements of NFPA 101, Chapter 38.3, limiting the common path of travel to less than 100 feet, dead-end corridors to 50 feet, and maximum travel distances to 300 feet.

2.2.7 Electrical/Communications

2.2.7.1 Power

2.2.7.1.1 Advanced Meter Reading System

To adhere to the Air Force Meter Data Management Plan, provisions are to be in place for a compliant electrical metering system. The metering at the electrical meter uses the necessary components for this system. Prior to commencement of new work, this system must be reviewed by qualified and licensed electrical personnel to confirm that the system meets the current and future application of the installation's Advanced Meter Reading System. To ensure properly networked communications, a telecommunications data outlet must be installed and connected near the electric meter to communicate with the centralized AMRS. Because of access constraints, this could not be confirmed at the time of visual inspection.

2.2.7.1.2 Stearley Heights Elementary School, B2261

Electrical panels are in electrical closets throughout the building. The electrical system appears to be in good physical condition and has no operational issues. The power equipment, which includes panels, transformers, and disconnects, is older than the recommended life of the equipment, with panel manufacture dates on or around 1984. The infrastructure consists of 208-volt and 480-volt panels and circuits.

Electrical infrastructure is required to be revised for any additional equipment added to the facility as a result of this effort, including new receptacles and circuits to support added users and workstations. At that time, a load study will be required to ensure existing panels and service can support new infrastructure.

2.2.7.1.3 Stearley Heights Gymnasium/Cafeteria, B2279

The electrical system that is installed in B2279 is in good physical condition and appears to have no issues. Power receptacles do not appear damaged. Electrical panels will require new permanent labels to meet code and installation labeling standards.

2.2.7.1.4 Utility Building, B2285

The electrical system for this building supports the power for control and operation of mechanical equipment. No revisions or upgrades are anticipated for this space.

2.2.7.2 Lighting Systems

2.2.7.2.1 Stearley Heights Elementary School, B2261

The building has recessed fluorescent light fixtures throughout each space. From a visual inspection, light levels appear adequate in most spaces. The lighting and controls should be updated throughout in accordance with UFC 3-530-01 and Illuminating Engineering Society of North America (IESNA) recommendations. Exterior lighting is generally limited to overhangs, canopies, and building entry doors. This lighting also should be upgraded. The new lighting includes code-complaint lighting controls and LED-type light fixtures.

In select spaces throughout the facility, emergency egress lighting does not meet emergency lighting requirements for National Electrical Code (NEC), IESNA, and NFPA. Additional emergency lighting fixtures or provisions are required to meet current codes and standards.

2.2.7.2.2 Stearley Heights Gymnasium/Cafeteria, B2279

The building has recessed fluorescent light fixtures throughout each space. From a visual inspection, light levels appear adequate in most spaces.

In general, emergency egress lighting does not meet emergency lighting requirements for NEC, IESNA, and NFPA. Additional emergency lighting fixtures or provisions are required to meet current codes and standards.

2.2.7.2.3 Utility Building, B2285

The building uses fluorescent strip lights to achieve its lighting. In general, the lighting in the building is inadequate and does not meet UFC 3-530-01 and IESNA recommendations. New lighting should include code-complaint lighting controls and LED-type light fixtures.

2.2.7.3 Communications Prewiring and Security

2.2.7.3.1 Stearley Heights Elementary School, B2261

The building contains an existing server room on the first level and one telecommunications rack on the second level. The facility serves fiber optic cabling to each of the telecom spaces. From each of the telecom rooms, the building distributes horizontal category 5e copper cabling to the work area outlets. The telecommunications room is not currently compliant with UFC 3-580-01 because this space is not dedicated and built out for network infrastructure distribution. The telecommunications rack on the second level also does not meet standards and must be given provisions to meet the UFC, such as security and environmental considerations.

Also contained in the building is the existing but abandoned phone cabling infrastructure. This includes copper terminal blocks mounted in recessed wall cabinets. The abandoned cabling, devices, and equipment must be removed from the facility to alleviate strain on available wall and ceiling space.

2.2.7.3.2 Stearley Heights Gymnasium/Cafeteria, B2279

The building has a telecommunications (telecom) wall cabinet serving the network infrastructure. The facility is served with copper cabling to the telecom cabinets. From the telecom cabinet, the building distributes horizontal copper cabling to the phone and work area outlets. There is no formal telecommunications space in this building. This does not comply with UFC 03-580-01 and Telecommunications Industry Association (TIA) standards.

Also contained in the building is the existing but abandoned phone cabling infrastructure. This includes copper terminal blocks mounted in recessed wall cabinets. The abandoned cabling, devices, and equipment must be removed from the facility to alleviate strain on available wall and ceiling space.

2.2.7.3.3 Utility Building, B2285

The facility does not contain formal telecommunications spaces to serve the building or network infrastructure. The telecommunications infrastructure and cabling do not meet current UFC 3-580-01 and TIA standards. As a utility space, this building requires a wall phone location, at a minimum, to achieve compliance with standards and for safety.

2.2.7.4 Lightning Protection and Grounding

A lightning protection system risk assessment should be performed in accordance NFPA 780 to determine if a lightning protection system should be provided or improved for this facility. Among other factors, the requirement for lightning protection is based on the facility occupancy, the requirement for the continuity of the facility services, the value of the facility contents, and the frequency of lightning strikes in Okinawa. The authority having jurisdiction (AHJ) will evaluate the risk assessment and requirements for lightning protection. The AHJ will have the final decision on whether it is necessary to equip a building with a compliant lightning protection system. If it is determined to be necessary, the lightning protection should be a conventional-type system using roof-mounted air terminals interconnected with surface-mounted roof conductors extended to a ground ring via down conductors. The ground rods for this system should be bonded to the ground ring using exothermic welds.

Test wells should be provided for a representative sample of the rods installed with one test well located near the main electrical switchboard or distribution panel.

All components of the interior and exterior distribution system should be bonded per host nation codes and the National Electrical Code (NEC). Static grounding should be provided for the telecommunications equipment racks and static dissipating floors in the telecommunications rooms. Surge protection should be provided for the main switchboard or main distribution panel and for panels serving sensitive electronic loads.

2.2.7.5 Real Property Installed Equipment Generators

This facility does not currently use real property installed equipment (RPIE) generators to back up the electrical service. The installation is expanding the power generation to back up much of the electrical distribution on Base.

2.2.8 Additional Mechanical, Electrical, and Plumbing Requirements

The building's new HVAC system must include a complete energy management and control system (EMCS) capable of communication with the Base EMCS. The system should have the capability of implementing advanced energy conservation algorithms such as pump and fan pressure reset and optimal start and stop. The system should operate as a fully automated direct digital control system to provide lower operating costs and ease of operation. The EMCS is to provide connection to the Base web control utility management control system front end located in Building 1474, allowing building control and monitoring from a central location. Utilities (water, electricity, hydronic, and other utilities as indicated in UFC 1-200-02) should be metered using AMRS protocols and be connected to the EMCS.

2.2.9 Antiterrorism

2.2.9.1 UFC 4-020-01, Security Engineering Planning

Specific antiterrorism/force protection measures relating to the site and the facilities should be specified in the contract documents to comply with UFC 4-010-01, DoD Minimum Antiterrorism Standards for Buildings. The Design Basis Threat (DBT) for these projects should be completed during the early stages of the design phase of each project. 718 CES should provide the final DBT in accordance with UFC 4-020-01, DoD Security Engineering Facilities Planning Manual.

Physical security requirements should be provided per Air Force Instruction (AFI) 31-101.

2.2.9.2 UFC 4-010-01, Antiterrorism Minimum Standards

For existing buildings, implementation of the UFC 4-010-01 standard is required for facilities with an inhabited occupancy and where the project renovation cost exceeds 50% of the replacement cost for the facility. Inhabited occupancy is defined as a building (or portion of building) routinely occupied by 11 or more DoD personnel and with a population density of greater than one person per 40 gross square meters (430 gross square feet). If the project costs do not exceed the 50% threshold, compliance is recommended, but not required. Additionally, facilities that do not meet the occupancy level for uninhabited are considered low-occupancy buildings and are exempt from the provisions of UFC 4-010-01.

B2285 is classified as low occupancy and does not require compliance. B2261 is classified as inhabited, but the renovation costs do not exceed 50% of the replacement cost, so compliance is recommended but not required. B2279 does not have any scope for this effort; therefore, compliance is not required.

2.2.10 Sustainability Requirements

The project should be designed/constructed to comply with the federal sustainability requirements as detailed in UFC 1-200-02, 1 December 2020 (Change 2, 1 June 2022), High Performance and Sustainable Building (HPSB)

Requirements; and the Air Force Sustainable Design and Development (SDD) Implementing Guidance Memorandum. Based on Table 1-1 of UFC 1-200-02, the project renovation cost is anticipated to not exceed 50% or more of the ERC and HPSB tracking and reporting is not required, and third-party certification is not required. However, tracking and reporting is recommended; therefore, the Air Force Sustainability Requirements Scoresheet should be used to track and report project compliance status at each submittal and is included in Appendix H.

UFC 01-200-02 does not apply to unoccupied buildings – those buildings occupied one hour or less per person per day on average. Therefore, utility building B2285 is not required to incorporate the UFC 01-200-02 requirements and will not require an Air Force Sustainability Requirements Scoresheet to track and report sustainable design initiatives.

Building B2279 is not scheduled to have any work performed; therefore, it does not have an Air Force Sustainability Requirements Scoresheet included.

The designer and construction contractor should provide the Air Force HPSB Checklist with each submission. Tracking and reporting includes revising and submitting the Air Force Sustainability Requirements Scoresheet and the supporting documents at each submittal until construction is complete.

2.2.11 Cybersecurity

Facility-related control system, as defined by the UFC as a subset of control systems that are used to monitor and control equipment and systems related to DoD real property facilities (building control systems, utility control systems, electronic security systems, and fire and life safety systems), should be upgraded in accordance with UFC 4-010-06.

The exact extent of the safeguards to be implemented are dependent on the system owner's determination of confidentiality, integrity, and availability impact levels for the system.

2.2.12 Airfields

Not applicable.

2.2.12.1 Geometrics and Grading

Not applicable.

2.2.12.2 Pavement

Not applicable.

2.2.13 Design Guidelines

In addition to information provided by the installation, the following references may serve as guidance for the planning and design of the project; however, this should not serve as a comprehensive list as more criteria may apply. Unless a specific document version or date is indicated, use criteria from the most current references, including any applicable addenda and changes, unless otherwise stated in the contract or task order, as of the date of the issue of the contract or task order. In the event of conflict between references and/or applicable criteria, apply the most stringent requirement, unless otherwise specifically noted.

- ABA Architectural Barriers Act Standards
- AFI 31-101 Security: Integrated Defense
- AFI 32-1021 Planning and Programming Military Construction (MILCON) Projects
- AFI 91-203 Air Force Consolidated Occupational Safety Instruction
- AFMAN 32-1062 Electrical Systems, Power Plants, and Generators

- AFMAN 32-1065 Grounding & Electrical Systems
- AFMAN 32-1084 Manual Facility Requirements
- Air Force Sundown Policy for Foam Fire Suppression Systems (Change 2, 29 December 2022)
- Air Force Sustainable Design and Development (SDD) Implementing Policy
- ASHRAE 90.1 Energy Standard for Buildings Except Low-Rise Residential Buildings
- ASHRAE Thermal Guidelines for Data Processing Environments
- ECB 2016-30 Air Force Sustainability Guidance for Third-Party Certification
- ECL Energy Conservation Law of Japan
- ETL 01-18 Fire Protection Engineering Criteria – Electronic Equipment Installations
- I3A Technical Criteria for the Installation Information Infrastructure Architecture
- IBC International Building Code
- Illuminating Engineering Society of North America (IESNA) Lighting Handbook
- JEMA Japanese Electrical Manufacturers' Association
- JIS Japanese Industrial Standards
- Kadena Air Base Architectural Compatibility Guide
- Kadena Air Base Design Guide
- NFPA 1 Fire Code
- NFPA 13 Standard for the Installation of Sprinkler Systems
- NFPA 20 Standard for the Installation of Stationary Pumps for Fire Protection
- NFPA 72 National Fire Alarm and Signaling Code
- NFPA 291 Recommended Practice for Water Flow Testing and Marking of Hydrants
- NFPA 70 National Electrical Code
- NFPA 70E Standard for Electrical Safety in the Workplace
- NFPA 101 Life Safety Code
- POC-TR 12-08 Standoff Distances for Japanese Conventional Construction
- UFC 1-200-01 DoD Building Code (General Building Requirements)
- UFC 1-200-02 High Performance and Sustainable Building Requirements
- UFC 3-101-01 Architecture
- UFC 3-110-03 Roofing
- UFC 3-120-01 Design: Sign Standards
- UFC 3-120-10 Interior Design
- UFC 3-190-06 Protective Coatings and Paints
- UFC 3-201-01 Civil Engineering
- UFC 3-201-02 Landscape Architecture
- UFC 3-210-10 Low Impact Development
- UFC 3-220-01 Geotechnical Engineering
- UFC 3-230-01 Water Storage and Distribution
- UFC 3-240-01 Wastewater Collection and Treatment
- UFC 3-250-01 Pavement Design for Roads and Parking Areas
- UFC 3-250-03 Standard Practice Manual for Flexible Pavements
- UFC 3-250-07 Standard Practice for Pavement Recycling
- UFC 3-301-01 Structural Engineering
- UFC 3-400-02 Design: Engineering Weather Data
- UFC 3-401-01 Mechanical Engineering
- UFC 3-410-01 Heating, Ventilating, and Air Conditioning Systems
- UFC 3-410-02 Direct Digital Controls for HVAC and Other Building Control Systems
- UFC 3-420-01 Plumbing Systems
- UFC 3-450-01 Noise and Vibration Control
- UFC 3-490-06 Elevators
- UFC 3-501-01 Electrical Engineering

- UFC 3-520-01 Interior Electrical Systems
- UFC 3-520-05 Stationary and Mission Batteries
- UFC 3-530-01 Interior and Exterior Lighting Systems and Controls
- UFC 3-540-01 Engine-Driven Generator Systems for Prime and Standby Power Applications
- UFC 3-550-01 Exterior Electrical Power Distribution
- UFC 3-570-01 Cathodic Protection
- UFC 3-575-01 Lightning and Static Electricity Protection Systems
- UFC 3-580-01 Telecommunications Interior Infrastructure Planning and Design
- UFC 3-600-01 Fire Protection Engineering for Facilities
- UFC 3-701-01 DoD Facilities Pricing Guide
- UFC 3-730-01 Programming Cost Estimates for Military Construction
- UFC 3-740-05 Handbook: Construction Cost Estimating
- UFC 4-010-01 DoD Minimum Antiterrorism Standards for Buildings
- UFC 4-010-06 Cybersecurity of Facility-Related Control Systems
- UFC 4-020-01 DoD Security Engineering Facilities Planning Manual
- UFC 4-021-01 Design and O&M: Mass Notification Systems
- UFC 4-021-02 Electronic Security Systems
- UFC 4-610-01 Administrative Facilities
- UFGS Unified Facility Guide Specifications
- USACE Japan District Design Guide
- Government of Japan Regulations Pertaining to Asbestos
    - Ambient Air Pollution Control Law
    - Waste Cleaning and Disposal Law
    - Asbestos Impediment Prevention Regulations
    - Industrial Safety and Health Law
    - Building Standards Law
- Ministry of Health, Labor, and Welfare Notification for Bulk Analysis of Building Material

3 Supporting Facilities

This section provides a conceptual-level narrative that describes identified requirements and notional design features or assumptions made for cost-estimating purposes. The DA is responsible for the final design solution to meet the project requirements.

3.1 Other User or Mission Facilities

Not applicable.

3.2 Site/Civil

3.2.1 General

Use of the existing facilities is to be maintained as each building is renovated. Therefore, revisions will not be made to the site unless required to meet the current UFCs.

3.2.2 Facility Demolition

No site demolition will be included. Most of the work is related to interior renovation.

3.3 Site Preparation

No site preparation should be required for the existing facilities. Staging and laydown areas will need to be identified for the sites. Space for staging and laydown areas exists south of B2261 either on the existing paved playground or on the turf athletic field.

3.4 Site Improvements

The required work for the facilities is located within each building. Minimal outside work is required to meet the current UFCs. Any work within the existing pavement will match the existing pavement.

3.4.1 Landscaping

Any required landscaping will be to stabilize disturbed areas.

3.5 Site Utilities Requirements

The facilities are existing and have utility services.

3.5.1 Water Supply

The potable water system for each building will remain as is. The fire protection water system is to be modified. Refer to the Fire Protection System discussion.

3.5.2 Sanitary Sewer

The sanitary sewer system is to remain as it currently exists.

3.5.3 Stormwater

The drainage patterns for the sites and the impervious/pervious areas will not change. No new stormwater management systems will be required.

3.6 Airfield Lighting and Navigational Aids

No airfield lighting or navigational aids are required for this renovation.

4 Equipment From Other Appropriations

4.1 User and Mission Requirements

Description	Quantity	Remarks (supporting utilities, supply source)
None anticipated	N/A	N/A

4.2 Equipment Details

4.2.1 Prewired Work Stations

Prewired workstations and other Furniture, Fixtures & Equipment (FF&E) will be government furnished, government installed. Final FF&E requirements should be developed as part of the Comprehensive Interior Design Package during the design of the project.

4.3 Information Systems Requirements

Active network communication equipment, located in telecommunications enclosures and spaces, is provided, programmed, and installed by Base communications personnel. These pieces of equipment, including network switches, are designed based on network users and mission requirements. Select provisions for wireless networks are under consideration to allow for more efficient use of the existing active network equipment and are to be connected to building network infrastructure via communications cabling. The need for additional coverage will be determined at a future time by the building users.

4.4 Non-RPIE Generators/Uninterruptible Power Supply/Other Backup Power Storage or Capability

None anticipated.

4.5 Other Equipment Items

None anticipated.

4.6 Raised Access Flooring

None anticipated.

4.7 TEMPEST/HEMP/RF Shielding

Not applicable.

4.8 Intrusion Detection System

None anticipated.

4.9 Closed Circuit TV

Building 2261 has a local video surveillance system that is not currently being used and not anticipated to be used in the future. It does not connect to the installation centralized security system. A new or upgraded electronic security system is not required at this time. However, the need for such systems should be reviewed at the time of project design and directed by installation security personnel based on installation standards. All components of electronic intrusion detection systems, including wiring and pathways, should meet UFC 3-580-01, UFC 4-021-02, and National Electric Code.

4.10 Other User Equipment

Not applicable.

5 Other Scope/Cost Considerations

5.1 Environmental Issues

Per Executive Order 12114, *Environmental Effects Abroad of Major Federal Actions*, Pacific Air Forces will follow the Environmental Impact Analysis Process in assessing the environmental impacts of the project. Minimal site work is anticipated for this effort; therefore, a soil screening survey should be performed during the design of the project to identify potential contaminants in the area. Environmental considerations include wetlands, archaeological or cultural sites, former contaminated sites, and unexploded ordnance.

The project will be required to comply with the following requirements during design and construction, which have been included for reference:

- A certified contractor is required to test for asbestos and lead-based paint (LBP) for suspect materials or for paint that will or may be disturbed in existing infrastructure.
- If applicable, submit contractor's sampling plan for review to SGXB and Civil, Environmental, and Infrastructure Engineering (CEIE). When sampling is complete, send results to SGXB and CEIE.
- Submit contractor's abatement plan to SGXB and CEIE if asbestos and/or LBP test results exceeding Japan Environmental Governing Standards will or may be disturbed.
- Submit safety data sheet for new paint products if scheduled to be used for this project.
- Ensure paint/primer used is low volatile and does not contain isocyanates or lead.
- The contractor is required to obtain prior coordination and approval from SGXB for use of any equipment containing radioactive material (RAM). Example: LBP XRF analyzer, soil compaction/density tester.
- Removal and disposal of lighting units and exit signs that may contain RAM requires SGXB coordination and approval. Coordinate with SGXB if they contain RAM prior to disposal.

Tables 5-1 and 5-2 highlight potential environmental concerns that may impact the successful execution of the project.

Table 5-1: Environmental Permits

Environmental Permits	Required Y/N/na	Remarks
Air Quality	na	
Water Quality	na	
Water Connection	N	
Wastewater	N	
Stormwater	N	
Solid Waste	N	
Hazardous Waste (RCRA)	na	
Environmental Construction	na	

Table 5-2: Hazardous Substance and Abatement Requirements

Hazardous Substances and Abatement Requirements	Required Y/N/na	Remarks
Asbestos	Y	Building and soil screening survey should be performed at each facility.
Lead-based Paint	Y	Building and soil screening survey should be performed at each facility.
Polychlorinated Biphenyls	Y	Building and soil screening survey should be performed at each facility.
Radon	na	
Installation Restoration Program Sites	na	
Toxic Industrial Waste	Y	Building and soil screening survey should be performed at each facility.
Radiological	na	
Heavy Metals	na	
Ozone Depleting Substances (refrigerants)	na	
Medical Gases	na	
Aqueous Film-Forming Foam	N	

5.2 Unexploded Ordnance, Cultural, and Real Estate Issues

Because the existing sites are disturbed, no issues are anticipated.

5.3 Site Accessibility

Base access is via the main gate (Gate 1) off Highway 58. Access to each site will require confirmation of the latest restricted/controlled area plan from Kadena AB to determine the extent of restricted area and escort requirements.

5.4 Construction Security

Perimeter controls should be maintained during all construction phases. Demolition and installation of the perimeter fence should be coordinated to maintain installation security. Also, coordinate with Base security if escorts should be required.

5.5 Construction Phasing

B2261 and B2285 are assumed to be available and turned over to the general contractor for the construction without phasing requirements.

5.6 Temporary User Space

B2261 cannot be occupied during the major renovation of the facility; therefore, temporary user space may be required depending on the tenants occupying the space prior to construction. Further coordination is required to determine the overall phasing plan of the tenants as the construction timeline is confirmed during design. At present, temporary user space or trailers are not included in the cost of the DD1391.

5.7 Approvals/Exemptions/Waivers

None anticipated.

[redacted]

6.1 Project Information

- Owner: Kadena Air Base
- Architect/Engineer: Jacobs
- Contracting Method: Design-Bid-Build

6.2 Project Construction Schedule

- Duration: 24 Months
- Start: July 2025 – Start dates are not set at this time
- Completion: July 2027
- Construction Midpoint for Escalation: July 2026

6.3 General Qualifications

- Estimate Basis
  - Layout: Programmed areas developed during charrette
  - Site Plan: Planning Charrette Report and out-brief during charrette
  - Sketches: Block drawings developed during charrette
  - Specifications: N/A
  - Narratives: Conceptual Charrette Report and out-brief during charrette
  - Requirements: Details from charrette; discussions with engineers/designers
- Report Format
  - The summary reports are presented to match the DD Form 1391 format
  - The estimates include an assembly level detail report that support the DD Form 1391 summary reports.
- Summary of Mark-ups
  - Prime contractor's general conditions, overhead, profit and bond
  - Subcontractor's overhead and profit
  - Escalation to midpoint of construction
  - 8 percent Supervision, Inspection and Overhead (SIOH)

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7 Appendices

Appendix A. Acronyms

[redacted]

Appendix D. Concept Drawings

Appendix E. Site and Facility Photos

[redacted]

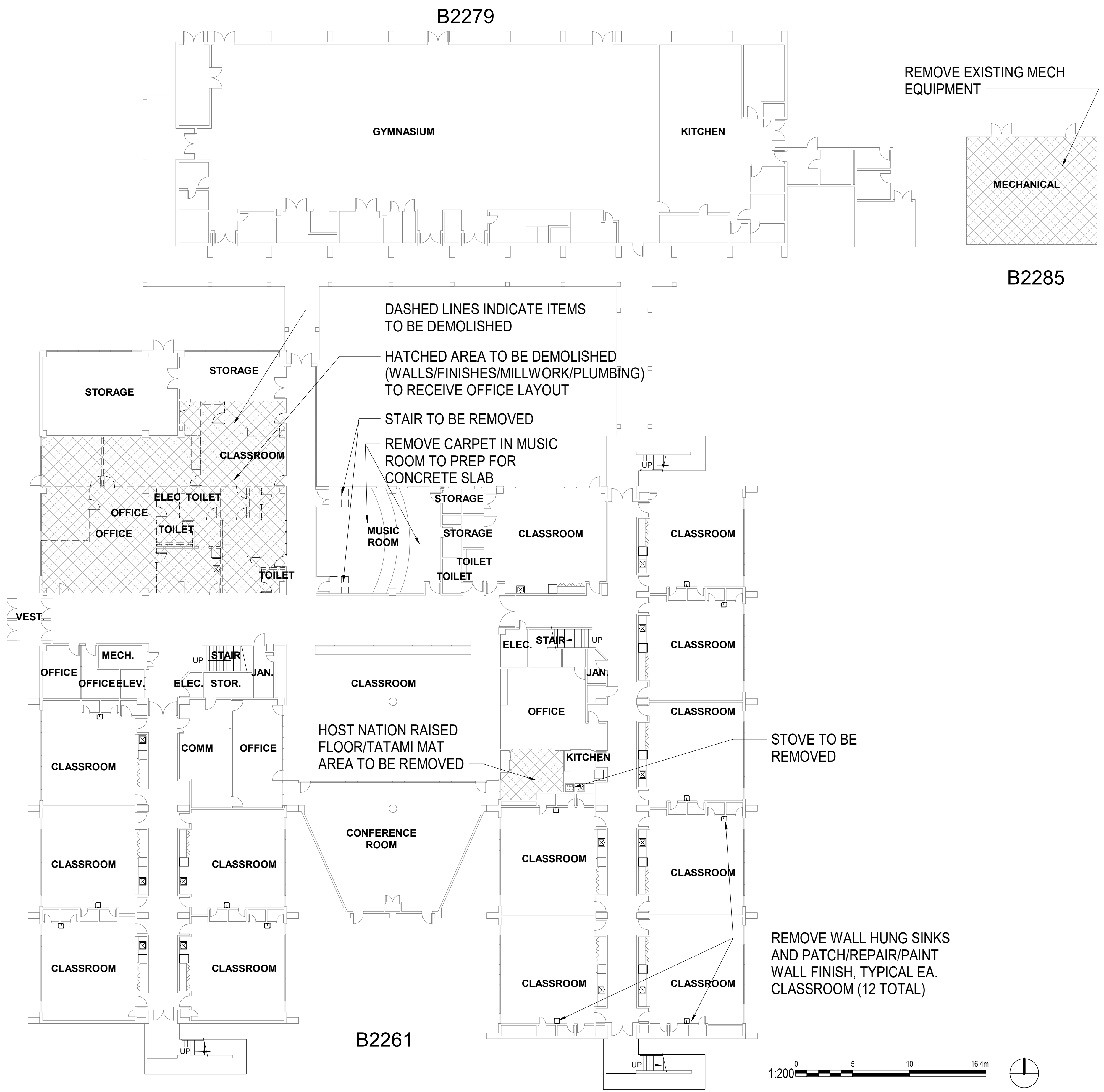
Appendix A Acronyms

Acronyms

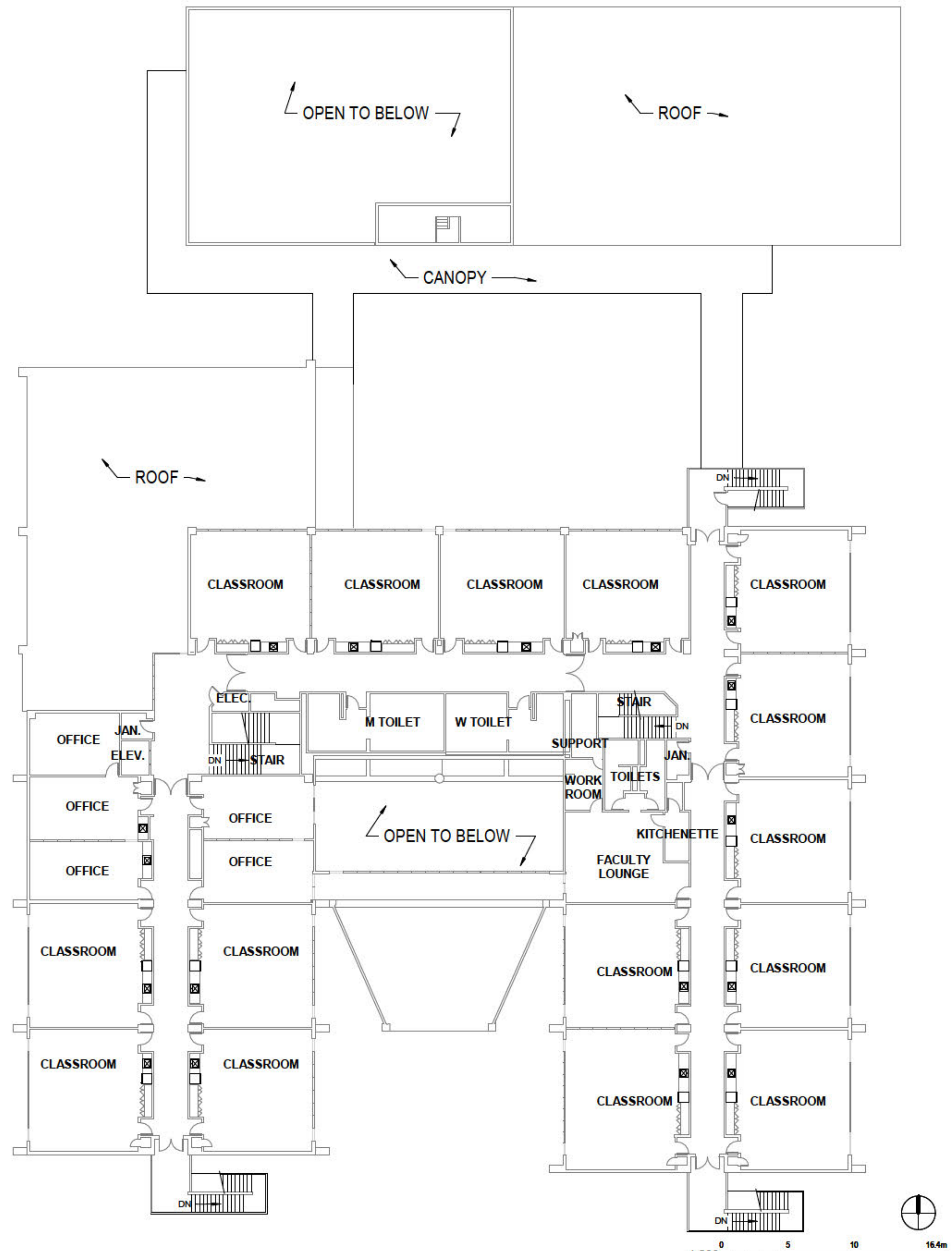
718 CES	718 Civil Engineer Squadron
AB	Air Base
ABA	Architectural Barriers Act
ACT	Acoustical Ceiling Tile
ADA	Americans with Disabilities Act
AFI	Air Force Instruction
AHJ	Authority having jurisdiction
AMCA	Air Movement and Control Association
AMRS	advanced meter reading system
ASCE	American Society of Civil Engineers
ASHRAE	American Society of Heating, Refrigeration, and Air-Conditioning Engineers
CATCODE	Category Code
CCR	Conceptual Charrette Report
CEIE	Civil, Environmental, and Infrastructure Engineering
CES	Civil Engineer Squadron
cfm	cubic foot (feet) per minute
COA	Course of Action
DA	Design Agent
DAFMAN	Department of the Air Force Manual
DBT	Design Basis Threat
DD	Department of Defense (form)
DOAS	dedicated outside air system
DoD	Department of Defense
EMCS	energy management and control system
ERC	estimated replacement cost
ESC	Environmental Severity Classification
FF&E	Furniture, Fixtures & Equipment
gpm	gallon(s) per minute
HPSB	High Performance and Sustainable Building
HVAC	heating, ventilation, and air conditioning
IBC	International Building Code
IESNA	Illuminating Engineering Society of North America
kPa	Kilopascal
LBP	lead-based paint
LCCA	life cycle cost analysis
LED	light-emitting diode
LOC	local operating console
LSC	Life Safety Code
MILCON	Military Construction
mm	millimeter(s)
MNS	mass notification system
NEC	National Electrical Code
NFPA	National Fire Protection Association
O&M	operations and maintenance
psf	pound(s) per square foot
psi	pound(s) per square inch
QFPE	qualified fire protection engineer
RAM	radioactive material
RPIE	real property installed equipment
SDD	Sustainable Design and Development
SF	square foot (feet)
TIA	Telecommunications Industry Association
UFC	Unified Facility Criteria
USACE	U.S. Army Corps of Engineers

[redacted]

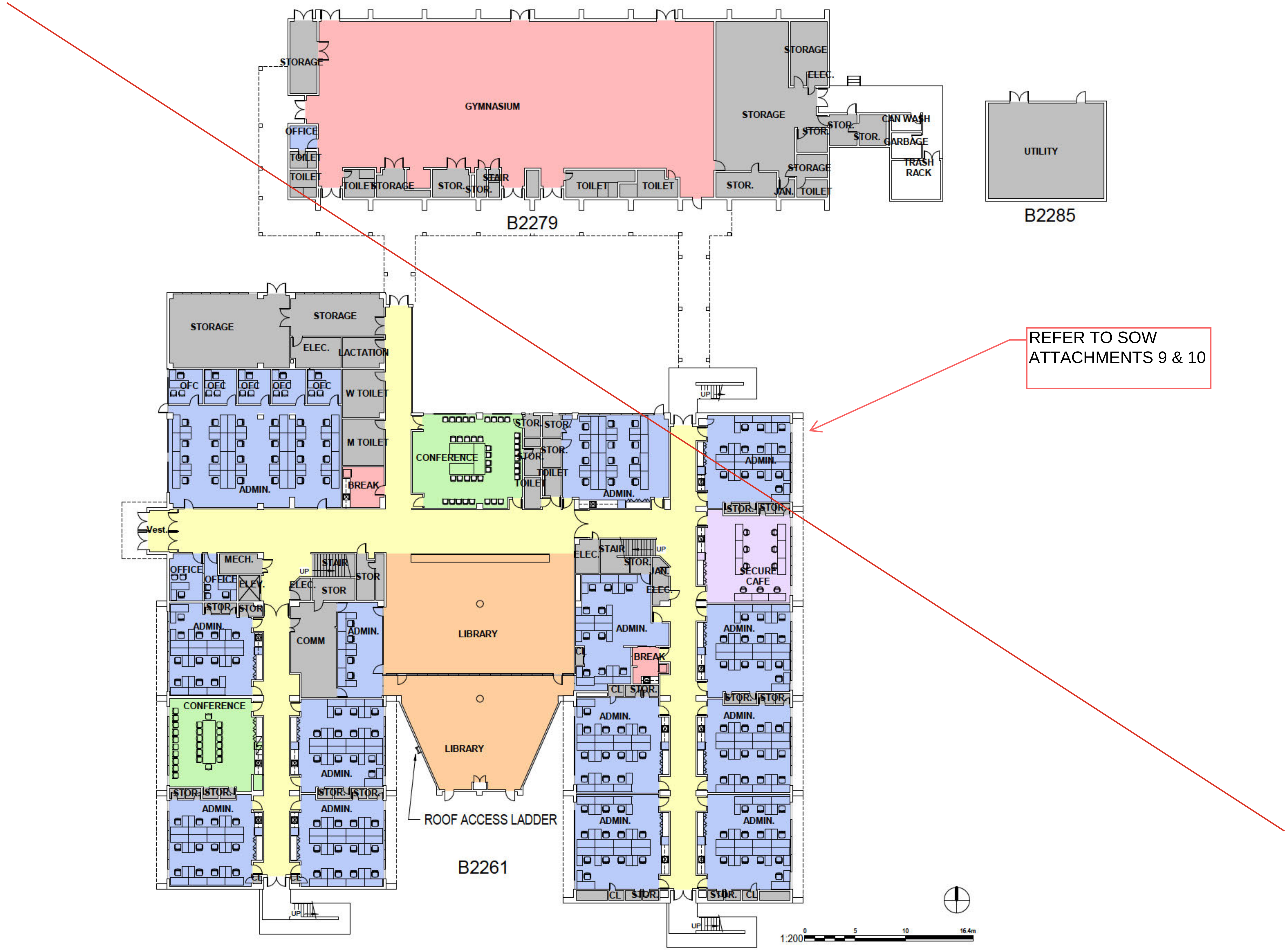
Appendix D Concept Drawings



CONCEPT EXISTING AND DEMOLITION PLAN - FIRST FLOOR

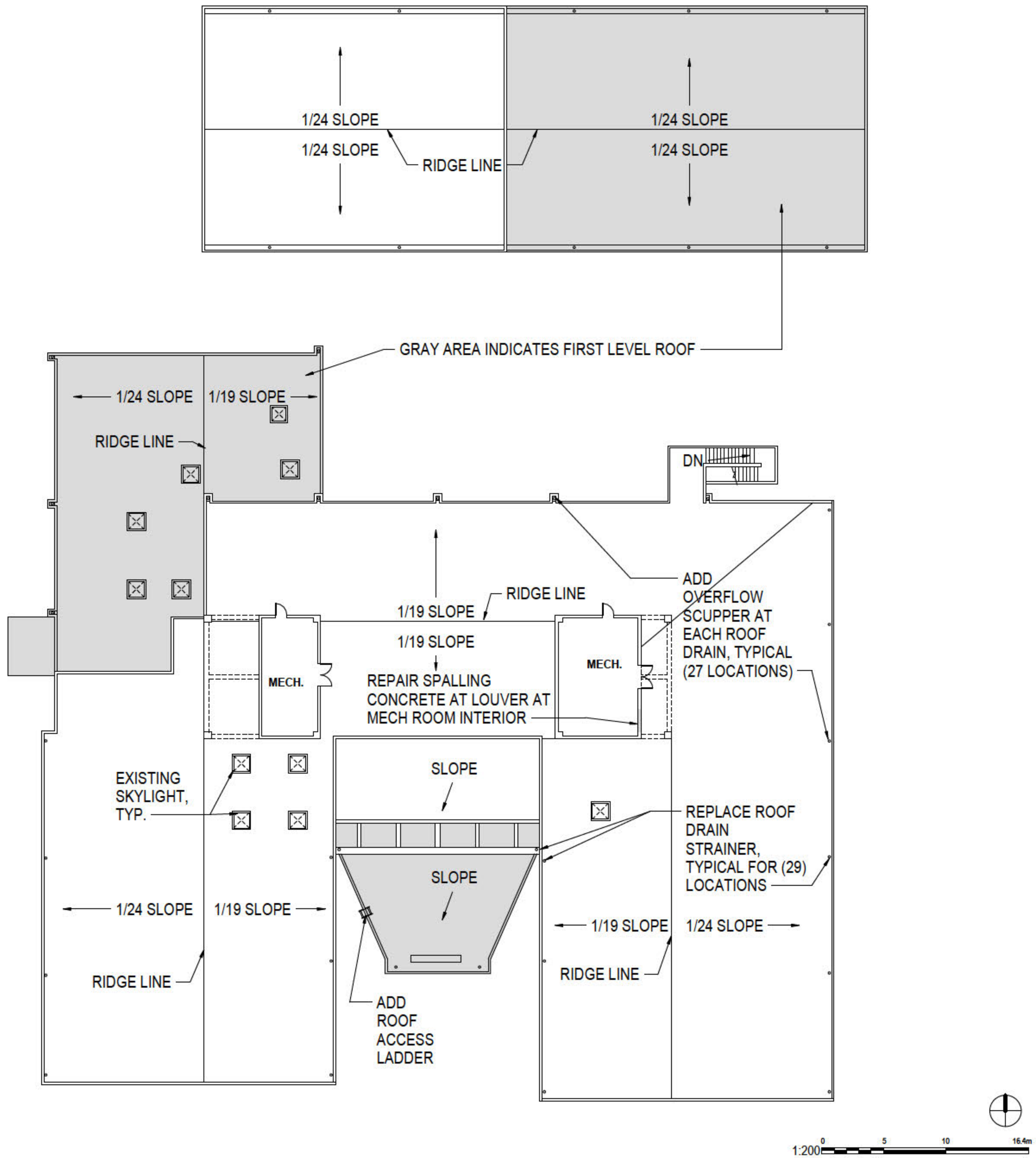


EXISTING PLAN - SECOND FLOOR



PREFERRED COA CONCEPT PLAN - FIRST FLOOR

PREFERRED COA CONCEPT PLAN - SECOND FLOOR



PREFERRED COA CONCEPT ROOF PLAN

Appendix E Site and Facility Photos

Exterior Photos B2261

Main entrance

Typical sunshade/façade for classroom wings

Exterior Photos B2261

View of south facade

North courtyard

Exterior Photos B2261

North façade/courtyard view

View along west facade

Exterior Photos B2261

View at 1996 addition connection (west side of addition)

View at 1996 addition connection (east side of addition)

Exterior Photos B2261

Typical window with exterior sunshade

Exterior Photos B2261

Roof access at northeast corner

Roof – view of west mechanical enclosure

Exterior Photos B2261

Interior view of spalling damage on east roof mechanical enclosure

Roof Drain – Note missing overflow scupper

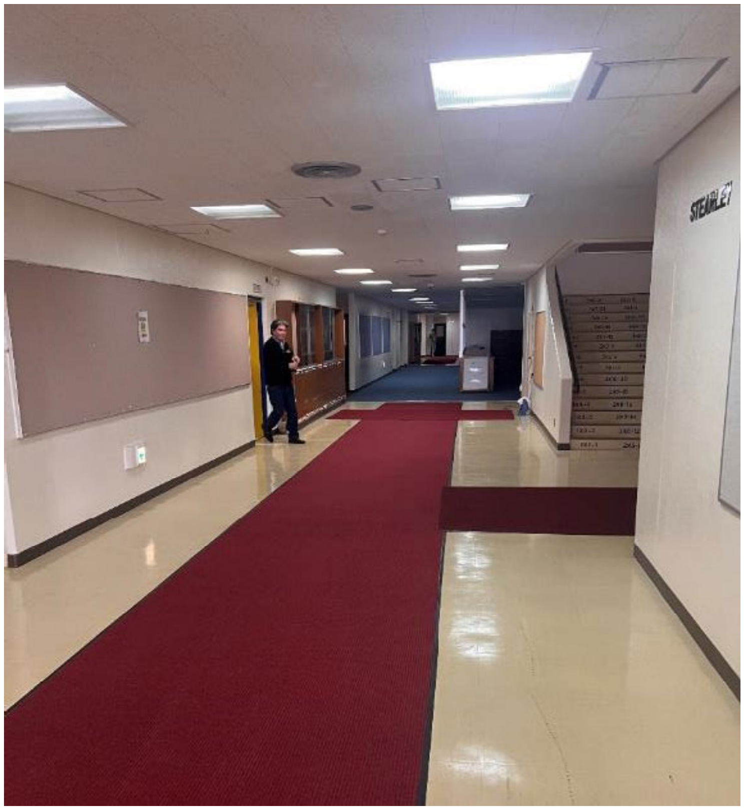
Roof drain – Corroded strainer

Exterior Photos B2261

View looking down to 1996 addition one-story roof

View looking down to 1996 addition one-story roof

Interior Photos B2261



Main entry corridor

Corridor view towards gymnasium exit

Interior Photos B2261

Administration break area – northwest quadrant

Break area at northwest quadrant administration area

Interior Photos B2261

Typical classroom – first floor

Typical classroom sink/refrigeration/storage cabinets – first floor

Interior Photos B2261

Typical classroom toilet

Typical classroom window

Interior Photos B2261

Central open area (future library)

Moveable partition at central area (future library)

Interior Photos B2261

Host Nation Tatami mat infill area

Host Nation Tatami mat infill area

Interior Photos B2261

Music room from entry door

Music room from back of classroom

Interior Photos B2261

Access panel in floor of Music room storage closet

East stairway

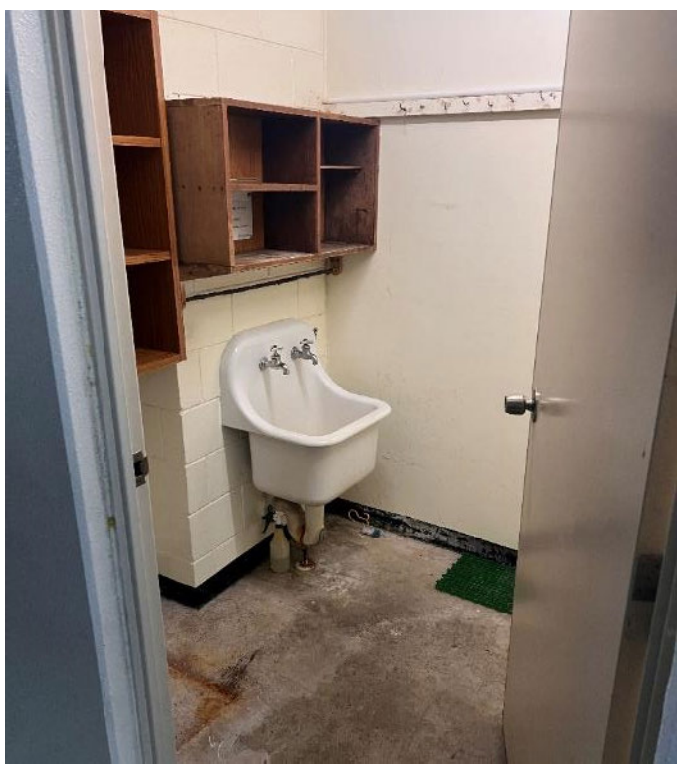
Interior Photos B2261

Second floor classroom – typical

Second floor classroom – typical

Interior Photos B2261

Water damage – second floor classroom



Janitor's closet

B2279 Gymnasium/Cafeteria

Exterior view of Gymnasium from courtyard

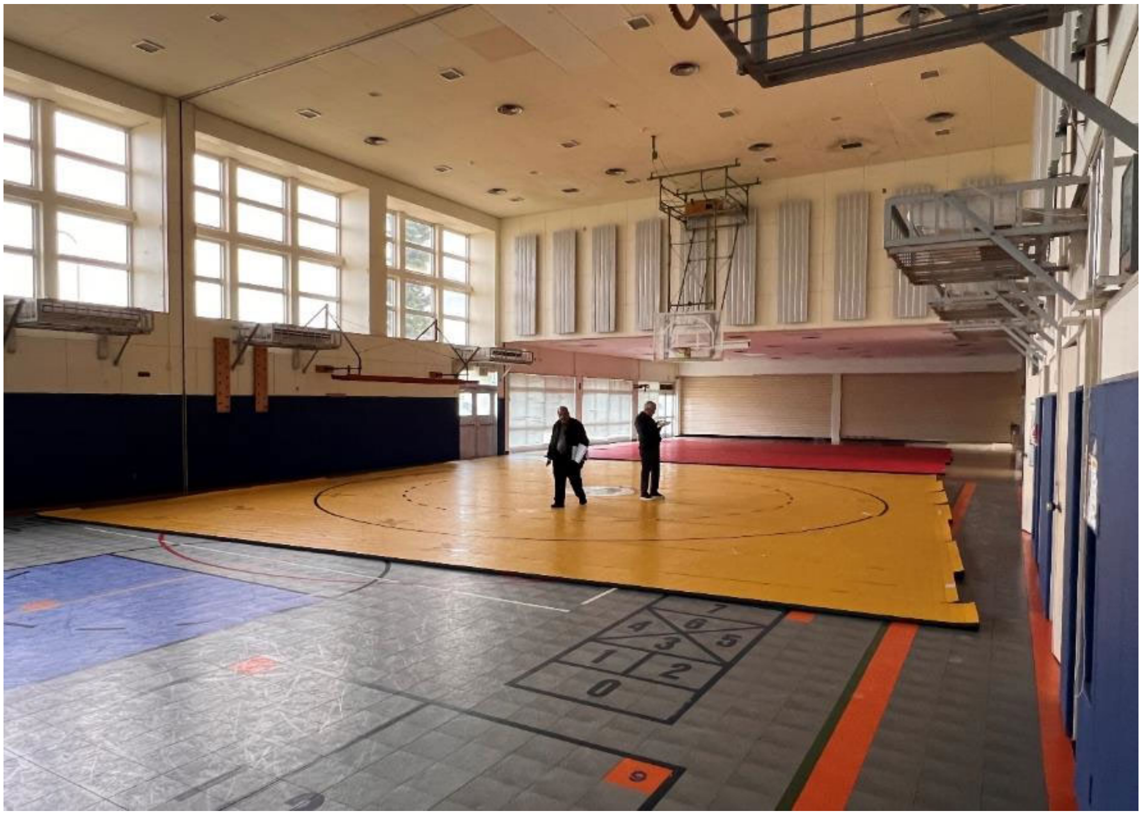
Gymnasium view from west side

B2279 Gymnasium/Cafeteria

Gymnasium view from courtyard

Trash and can wash area for cafeteria

B2279 Gymnasium/Cafeteria



Gymnasium view northeast

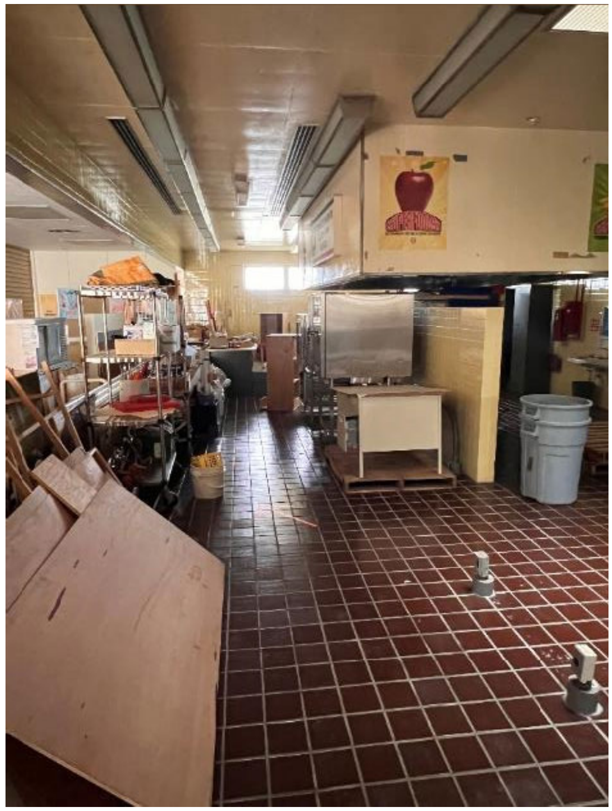
Gymnasium view west

B2279 Gymnasium/Cafeteria

Gymnasium HVAC units

Gymnasium entry doors

B2279 Gymnasium/Cafeteria



Cafeteria/kitchen area

Cafeteria/kitchen area

B2285 Utility Building

B2285 Exterior view from B2261 roof

B2285 entry doors

B2285 Utility Building

Mech/plumbing equipment

Mech/plumbing equipment

B2285 Utility Building

Mech/plumbing equipment

Mech/plumbing equipment

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